RLVAE – Full Method

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This document defines the **complete scientific structure** of the Riemannian Latent VAE (RLVAE) method, designed to be exhaustive, mathematically rigorous, and directly suitable for a journal or thesis chapter. It is inspired by and extends the current experimental methodology and notes, but reorganised for full clarity and reviewer-grade precision.

PART I — Global Framework & Problem Formulation

This section introduces the conceptual, probabilistic, and geometric foundations of the Riemannian Latent Variational Autoencoder (RLVAE). We precisely define the modelling problem, the nature of the data, and the motivations for embedding the latent space within a learned Riemannian manifold, highlighting the limitations of classical Euclidean VAEs for structured longitudinal phenomena.

1.1 Objective and Scope

The objective of this work is to design and analyse a generative latent variable model capable of representing, reconstructing, and interpolating longitudinal data that evolve along structured, low-dimensional, and non-linear trajectories. In particular, we focus on scenarios where the evolution of observations is governed by underlying continuous processes (e.g. biological growth, disease progression, morphological deformation) that naturally give rise to curved manifolds in latent space.

Standard Variational Autoencoders (VAEs) assume an isotropic Euclidean latent space endowed with a fixed Gaussian prior, typically $p(z) = \mathcal{N}(0, I)$.
While effective for static and unstructured datasets, this assumption imposes a geometry that is incompatible with the intrinsic structure of many real-world temporal processes. In such settings, Euclidean models fail to capture:

- anisotropic variability across latent directions,
- non-uniform data density,
- preferred evolutionary paths or constrained trajectories.

The RLVAE framework addresses these limitations by endowing the latent space with a learned Riemannian structure that explicitly models local anisotropy, curvature, and density variation. This geometry governs not only the organisation of latent representations, but also the dynamics of sampling, trajectory interpolation, and uncertainty propagation.

The method is organised as a three-stage pipeline:

- **Stage A:** Training of a standard VAE to obtain a stable initial latent embedding.
- **Stage B:** Extraction and learning of a Riemannian metric encoding local geometric structure.
- **Stage C:** Construction of a Riemannian VAE integrating Hamiltonian dynamics and normalizing flows guided by the learned metric.

The present document focuses on the complete mathematical formulation of Stage C, while maintaining strict coherence with the metric and embedding produced in earlier stages.

1.2 Data Model and Longitudinal Structure

Let $x \in \mathcal{X}$ denote an observed data point, typically represented as a grayscale image. In the longitudinal setting, each subject i is associated with a sequence:

$$\mathbf{x}_i = x_i^{(1)}, x_i^{(2)}, \ldots, x_i^{(T)}$$

where T denotes the number of temporal observations.

To isolate the geometric properties of the model under controlled conditions, we adopt a synthetic dataset based on sequences of evolving ellipses.
Each sequence follows a smooth and deterministic morphological evolution in time, ensuring:

- continuous temporal deformation,
- low intrinsic dimensionality,
- absence of confounding nuisance factors.

The data generative mechanism is constrained to a small number of degrees of freedom, typically:

- major radius (size),
- eccentricity (shape deformation),
- optionally orientation or translation.

As a result, the true data-generating process lies on a smooth and low-dimensional manifold embedded in the high-dimensional image space. The modelling objective is therefore not simply reconstruction, but the recovery of a latent space whose geometry faithfully reflects this intrinsic structure, enabling meaningful interpolation, temporal extrapolation, and dynamically consistent trajectory modelling.

1.3 Probabilistic Generative Model (Longitudinal Formulation)

We consider a generative latent variable model explicitly designed for longitudinal data.
We denote by the superscript j the subject (patient) index and by the subscript $t \in \{0, \ldots, T\}$ the temporal index.

For each subject j, we observe a temporally ordered sequence:

$$\mathbf{x}^j = \{x_0^j, x_1^j, \ldots, x_T^j\},$$

with corresponding latent variables:

$$\mathbf{z}^j = \{z_0^j, z_1^j, \ldots, z_T^j\},$$

where each latent state $z_t^j \in \mathcal{M}$ lies on a smooth differentiable manifold endowed with a Riemannian metric $G(z)$.

The joint generative model for a single subject trajectory is defined as:

$$p_\theta(\mathbf{x}_{0:T}^j, \mathbf{z}_{0:T}^j) = p(z_0^j) \prod_{t=0}^{T} p_\theta(x_t^j \mid z_t^j) \prod_{t=1}^{T} p(z_t^j \mid z_{t-1}^j),$$

where:

- $p_\theta(x_t^j | z_t^j)$ denotes the decoder likelihood at time t,
- $p(z_0^j)$ is the initial latent prior,
- $p(z_t^j | z_{t-1}^j)$ defines the temporal latent dynamics.

In the RLVAE framework, temporal evolution is not modelled as an additive stochastic transition but as a deterministic geometric transport governed by the learned Riemannian structure.
Specifically, we define:

$$z_t^j = f_t(z_{t-1}^j), \qquad t = 1, \ldots, T,$$

where each $f_t$is a diffeomorphic flow constrained by the metric tensor.
This induces for each subject a latent trajectory $\{z_t^j\}_{t=0}^T$ that follows the intrinsic geometry of the manifold.

Under this formulation, the prior over the full latent trajectory of subject j is implicitly induced by the prior on the initial state and the deterministic flow sequence:

$$p(\mathbf{z}_{0:T}^j) = p(z_0^j) \prod_{t=1}^{T} \delta\big(z_t^j - f_t(z_{t-1}^j)\big)$$

Since the posterior distribution over full trajectories is intractable, it is approximated by a variational distribution:

$$q_\phi(\mathbf{z}_{0:T}^j \mid \mathbf{x}_{0:T}^j),$$

leading to the sequence-level Evidence Lower Bound (ELBO) for subject j:

$$\mathcal{L}(\theta, \phi; \mathbf{x}_{0:T}^j) = \sum_{t=0}^{T} \mathbb{E}_{q_\phi(z_t^j | \mathbf{x}_{0:T}^j)}\big[\log p_\theta(x_t^j | z_t^j)\big] - \mathrm{KL}\big(q_\phi(\mathbf{z}_{0:T}^j | \mathbf{x}_{0:T}^j) \parallel p(\mathbf{z}_{0:T}^j)\big).$$

In contrast to classical VAEs, which assume temporally independent latent variables, the RLVAE explicitly models each subject's latent trajectory as a structured geometric object transported through time via flow-based dynamics.
This formulation enables coherent interpolation, prediction, and analysis of longitudinal sequences while preserving both temporal consistency and geometric fidelity.

This establishes a principled probabilistic foundation for subject-wise sequence modelling on Riemannian manifolds, forming the theoretical basis for the posterior construction developed in Stage C.

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2 - Latent Geometry Construction (Stage B)

Stage B takes the Euclidean latent space learned in Stage A and upgrades it into a **Riemannian manifold**.
Starting from the aggregate posterior codes of a vanilla VAE, we construct a position-dependent metric tensor $G(z)$ that encodes how space should be **stretched**, **compressed**, and **curved** to reflect the **structure of the data**.

This learned geometry plays three roles simultaneously:

- It defines **distances and geodesics** in latent space, shaping how trajectories interpolate between codes.
- It induces a **Riemannian volume form**, from which we build a geometry-aware prior.
- It governs the **Riemannian Hamiltonian dynamics** used in Stage C to refine the posterior and generate temporally coherent paths.

Unlike standard geometric VAEs, which directly identify the metric with a precision field, we adopt a **Precision-First** viewpoint: we first learn a latent **precision field** $(P(z)$(information density), then define the metric as its inverse $G(z) = P(z)^{-1}$.
This inversion is what makes latent trajectories flow through high-density regions rather than around them.

In this section, we formalise the construction of $P(z)$ and $G(z)$, analyse their **spectral** and **curvature** properties, and explain how they are coupled to the variational encoder.

2.1 Metric Learning Philosophy and Problem Statement

We assume the existence of a pre-trained probabilistic encoder from Stage A, which maps observations to a Euclidean latent space.
Stage B aims to endow this space with a Riemannian structure derived solely from the aggregate posterior distribution.

> **Definition 2.1 - The Empirical Latent Distribution**
>
> Let $\mathcal{X}$ be the data space and $\mathcal{Z} \cong \mathbb{R}^d$ be the latent manifold.
> Let $q_\phi^{(A)}(z|x) = \mathcal{N}(z; \mu^{(A)}(x), \Sigma^{(A)}(x))$ denote the variational posterior of the pre-trained Stage A VAE.
> We define the **Empirical Latent Support** $\mathcal{D}_Z$ as the set of posterior means aggregated over the training dataset $\mathcal{D} = \{x_i\}_{i=1}^N$:
>
> $$\mathcal{D}_Z = \left\{ \mu_i \in \mathcal{Z} \mid \mu_i = \mu^{(A)}(x_i),\, x_i \in \mathcal{D} \right\}$$

Remark: In this framework, we treat the set $\mathcal{D}_Z$ as the discrete skeleton of the data manifold.
The variance $\Sigma^{(A)}$ from Stage A is discarded for the purpose of geometry construction, as it reflects the isotropic Euclidean assumption we wish to replace.

2.1.1 The Precision-First Hypothesis

Standard geometric VAEs typically model the metric tensor $G(z)$ directly.
However, this leads to conceptual difficulties when modeling "empty" space, where distance should be large (high G) but information is low.
To address this, we introduce the **Precision-First paradigm**.

> **Hypothesis 2.1 - Precision-First Manifold**
>
> The intrinsic geometry of the latent space is generated by a scalar field of Information Density, represented by the precision matrix $P(z)$.
> The Riemannian metric tensor $G(z)$ is defined strictly as the inverse of this precision field:
>
> $$G(z) \equiv P(z)^{-1}$$

Under this hypothesis:

- **High Density Regions** (Data Clusters) are characterized by high precision ($P \succ 0$ is large), implying low metric cost (G is small).
- **Low Density Regions** (Voids) are characterized by low precision ($P \to 0$), implying high metric cost (G is large).

Important Convention – Metric vs Precision

Throughout this work, we explicitly distinguish between:

- The **Precision Field** $P(z)$, encoding local information density,
- The **Riemannian Metric Tensor** $G(z)$, defining geometric cost.

They are related by the strict inversion:

$$G(z) = P(z)^{-1}$$

All geometric operators (distances, geodesics, Hamiltonian dynamics) are derived from $G(z)$, while probabilistic concentration and prior weighting are governed by $P(z)$.
Confusing these two quantities leads to fundamentally **different manifold behaviour** and sampling dynamics.

2.1.2 Structural Constraints

To ensure the learned geometry is topologically stable and computationally tractable, we impose the following structural constraints on the precision field $P(z)$.

Definition 2.2 - Centroid-Based Parameterization

We approximate the underlying manifold structure using a finite set of geometric anchors.
The precision field is parameterized by a tuple $\mathcal{G} = \{\mathbf{C}, \mathbf{\Lambda}, T, \lambda\}$, where:

- $\mathbf{C} = \{c_k\}_{k=1}^{K} \subset \mathbb{R}^d$ is a set of K learnable **centroids**.
- $\mathbf{\Lambda} = \{\Lambda_k\}_{k=1}^{K} \subset \mathbb{S}_{++}^d$ is a set of symmetric positive-definite (SPD) **local precision matrices**.
- $T \in \mathbb{R}^+$ is a **temperature** scalar controlling the smoothness of the manifold.
- $\lambda \in \mathbb{R}^+$ is a **regularization floor**.

The construction of the continuous field $P(z)$ from these discrete parameters is detailed in Section 2.2.

2.2 Formal Construction of the Metric Field

Given the discrete parameter set $\mathcal{G} = \{\mathbf{C}, \mathbf{\Lambda}, T, \lambda\}$, we construct a continuous Riemannian manifold $\mathcal{M} = (\mathcal{Z}, G)$ by interpolating the local structural information.

Definition 2.3 - The Interpolation Kernel

We define a smooth weighting function $w : \mathcal{Z} \to \Delta^{K-1}$ that maps any latent point z to a probability simplex over the K centroids. The weight of the k-th centroid at location z is given by the softmax of the negative squared Euclidean distances scaled by the temperature T:

$$w_k(z) = \frac{\exp\left(-\frac{\|z-c_k\|_2^2}{T}\right)}{\sum_{j=1}^{K} \exp\left(-\frac{\|z-c_j\|_2^2}{T}\right)}$$

Remark: The temperature T controls the locality of the manifold structure:

- As $T \to 0$, the geometry becomes piecewise constant (Voronoi tessellation).
- As $T \to \infty$, the geometry approaches a global average of all centroids.

Definition 2.4 - The Precision Field

We define the scalar field of Information Density, represented by the Precision Matrix $P(z)$, as the regularized convex combination of the local centroid precisions:

$$P(z) = \lambda I_d + \sum_{k=1}^{K} w_k(z)\Lambda_k$$

where I_d is the $d \times d$ identity matrix and $\lambda > 0$ is the regularization floor.

Proposition 2.1 - Global Validity

For any $z \in \mathcal{Z}$, the precision matrix $P(z)$ is **Symmetric Positive Definite** (**SPD**).

Proof:

By definition, each local precision matrix Λ_k is **SPD** ($\Lambda_k \in \mathbb{S}^d_{++}$), meaning $v^\top \Lambda_k v > 0$ for all non-zero vectors v.
The weights $w_k(z)$ are strictly positive and sum to 1. The term λI_d is also SPD for $\lambda > 0$.
Since the set of **SPD** matrices is a *convex cone*, any convex combination of **SPD** matrices plus an **SPD** regularization term is strictly SPD.
Thus, $P(z) \succ 0$ for all z.

Definition 2.5 - The Riemannian Metric Tensor

Consistent with Hypothesis 2.1 (**Precision-First**), the **Riemannian metric tensor** $G(z)$ is defined explicitly as the inverse of the precision field:

$$G(z) \equiv P(z)^{-1} = \left(\lambda I_d + \sum_{k=1}^{K} w_k(z)\Lambda_k\right)^{-1}$$

This metric tensor varies smoothly with z and defines the local geodesic cost function.

Corollary 2.1 - Metric Bounds

The spectrum of the metric tensor is **bounded**.
Let $\lambda_{\min}(\Lambda_k)$ and $\lambda_{\max}(\Lambda_k)$ be the minimum and maximum eigenvalues of the centroid precisions.
The eigenvalues of the metric $G(z)$ are bounded by:

$$\frac{1}{\lambda + \max_k \lambda_{\max}(\Lambda_k)} \le \lambda_i(G(z)) \le \frac{1}{\lambda}$$

Proof:

The precision $P(z)$ is bounded above by the maximum possible weighted sum plus λ, and below by λ.
Since $G = P^{-1}$, its eigenvalues are the reciprocals of P's eigenvalues, reversing the inequalities.
This guarantees that the metric never becomes singular (collapse) or infinite (explosion).

Remark (Centroid Approximation Justification):

The centroid-based formulation provides a **sparse** and **interpretable** approximation of the latent geometry, avoiding the instability of kernel density estimators in high dimensions.
It ensures **linear complexity** in the number of centroids and allows explicit control over smoothness and conditioning.
This structural choice enables **scalable learning** while preserving the **essential topology** of the data manifold.

2.3 Spectral Properties and Geometric Stability

The stability of the Riemannian Hamiltonian dynamics (Stage C) depends critically on the spectral properties of the metric tensor.
We formally characterize the role of the regularization parameter λ in bounding the manifold's deformation.

Definition 2.6 - **The Background Geometry**

The regularization term λI_d in the precision field $P(z)$ induces a **Background Euclidean Geometry.**
In the limit $\|z\| \to \infty$ (far from all centroids), the weights $w_k(z) \to 0$ (assuming bounded centroids), and the manifold converges to a flat Euclidean space with constant metric:

$$\lim_{\|z\|\to\infty} G(z) = (\lambda I_d)^{-1} = \frac{1}{\lambda} I_d$$

Remark: This implies that the "void" (empty space) is not undefined; it is a scaled Euclidean space where distances are large (since λ is typically small, $1/\lambda$ is large).
This provides a restorative force during sampling: random walks in the void become energetically expensive Euclidean random walks.

Proposition 2.2 - **Bounded Condition Number**

The Riemannian metric $G(z)$ is globally well-conditioned. Its condition number $\kappa(G(z))$ is strictly bounded.

Proof:
Let $\Lambda_{max}^{\sup} = \max_k \lambda_{\max}(\Lambda_k)$ be the maximum eigenvalue across all centroid precision matrices.
From Corollary 2.1, the eigenvalues of the precision $P(z)$ lie in $[\lambda, \lambda + \Lambda_{max}^{\sup}]$.
Since $G(z) = P(z)^{-1}$, the condition number of the metric is identical to the condition number of the precision:

$$\kappa(G(z)) = \frac{\lambda_{\max}(P(z))}{\lambda_{\min}(P(z))} \leq \frac{\lambda + \Lambda_{max}^{\sup}}{\lambda} = 1 + \frac{\Lambda_{max}^{\sup}}{\lambda}$$

This upper bound is constant and independent of z.
→ The manifold curvature cannot become arbitrarily sharp, and the metric cannot degenerate, guaranteeing the numerical stability of the leapfrog integrator used in Stage C.

Definition 2.7 - **Manifold Smoothness**

The smoothness of the geometry is controlled by the temperature T.
The gradient of the metric field $\nabla_z G(z)$ is bounded for $T > 0$.

- **Low T:** Transitions between centroid-dominated regions are sharp, increasing local curvature and potentially creating **metric walls**.
- High T: Transitions are diffuse, flattening the manifold towards the global average precision.

We select T to balance the fidelity of the local structure against the ease of traversal by the sampler.

2.4 Differential Structure and Local Curvature

The Riemannian metric field $G(z)$ defines not only distances but also the local curvature of the latent manifold.
This differential structure governs how trajectories bend and how the Hamiltonian dynamics deviate from Euclidean paths.

Proposition 2.3 - **Differential Regularity**

For any temperature $T > 0$, the metric tensor field $G(z)$ is C^∞ over $\mathcal{Z}$.

Proof:
The weighting functions $w_k(z)$ are composed of exponential and normalization operations, which are smooth everywhere.
The precision field $P(z)$ is a linear combination of constant matrices weighted by $w_k(z)$, inheriting smoothness.
Since $P(z)$ is strictly positive definite (Proposition 2.1), the matrix inverse operation $G(z) = P(z)^{-1}$ is smooth.
Consequently, all partial derivatives $\partial G_{ij}/\partial z_k$ exist and are continuous.

Definition 2.8 - The Levi-Civita Connection

The **curvature** of the manifold is encoded by the Christoffel symbols of the second kind, derived from the metric derivatives:

$$\Gamma_{ij}^{k}(z) = \frac{1}{2} G^{kl}(z) \left(\frac{\partial G_{lj}}{\partial z_i} + \frac{\partial G_{li}}{\partial z_j} - \frac{\partial G_{ij}}{\partial z_l} \right)$$

These symbols define the **Geodesic Equation**, which governs the unforced motion of a particle on the manifold:

$$\ddot{z}^k + \Gamma_{ij}^{k}(z) \dot{z}^i \dot{z}^j = 0$$

Corollary 2.2 - Curvature Control

The Riemannian curvature tensor is **finite everywhere** for $T > 0$.
As $G(z)$ and its derivatives are bounded, the sectional curvature remains **controlled**.
No singular geometric walls (infinite curvature) emerge unless $T \to 0$.

Remark (Interpretation of Temperature):
The temperature T acts as a regularization hyperparameter for curvature:

- **Low T:** Induces sharp transitions between centroid-dominated regions. This creates ***metric ridges*** with high curvature, which can destabilize numerical integrators.
- **High T:** Flattens the curvature, causing the geometry to approach a global Euclidean average.

We select T to ensure that latent trajectories remain dynamically stable while preserving structurally significant curvature.

2.5 Geometric Interpretation and Probabilistic Coupling

The metric field $G(z)$ provides the structural link between the latent geometry and the probabilistic generative model.
We now formalize how this metric dictates volume measurement, geodesic transport, and the local uncertainty quantification.

Definition 2.9 - Riemannian Volume and Information Density

The Riemannian volume element associated with the metric $G(z)$ is given by $dV_G(z) = \sqrt{\det G(z)}\, dz$.
Under the Hypothesis 2.1 (**Precision-First**), we identify the determinant of the precision matrix as a proxy for **Information Density**:

$$\mathrm{Info}(z) \equiv \det P(z) = \det G(z)^{-1} = \frac{1}{\det G(z)} \quad \text{and} \quad d\mu(z) \propto \sqrt{\det P(z)} dz$$

Here, $\det P(z)$ acts as a **proxy** for local information concentration, while the **geometric prior weight** is proportional to its square root, consistent with **Riemannian volume theory**.

Interpretation:

- **High-Information Regions** (Data Clusters): The precision $P(z)$ is large (dominated by Λ_k).
  Consequently, $\det G(z)$ is **small** → The metric locally **compresses** space, assigning small geometric volume to the region.
- **Low-Information Regions** (The Void): The precision $P(z) \to \lambda I$.
  Consequently, $\det G(z)$ is **large** → The metric locally **expands** space, assigning large geometric volume to the empty regions.

Proposition 2.4 - **Geodesic Energy Cost**

Latent trajectories **minimize** the Riemannian energy functional.
This implies that geodesic paths preferentially traverse **high-information regions** (low metric cost), resulting in naturally data-aligned trajectories.
In contrast to **standard geometric VAEs**, where trajectories tend to diverge toward **low-density regions**, the inverted precision formulation ensures that geodesics are attracted toward the empirical data manifold.
Regions of **high information** density act as **conductive channels** for geodesic transport.

Proof:
The energy of a curve $\gamma : [0, 1] \to \mathcal{Z}$ is defined as:

$$\mathcal{E}(\gamma) = \int_0^1 \dot{\gamma}(t)^\top G(\gamma(t))\,\dot{\gamma}(t)\,dt = \int_0^1 \dot{\gamma}(t)^\top P(\gamma(t))^{-1}\,\dot{\gamma}(t)\,dt$$

In regions of high information density ($P(z)$ is large), the inverse $P(z)^{-1}$ is small (in the Loewner order).
Therefore, the energetic cost of traversing these regions is low.
Conversely, in voids ($P(z)$ is small), the cost $G(z)$ is high.
Thus, geodesics and Hamiltonian trajectories are **naturally confined** to the high-density manifold structure, penalizing excursions into the void.

Definition 2.10 - **The Geometrically Consistent Covariance**

To couple the variational encoder with the learned geometry, we define the local posterior $q_\phi(z|x)$ as a **Riemannian Normal Distribution**.
In information geometry, the metric tensor $G(z)$ corresponds to the **Fisher Information Matrix**, which acts as the precision matrix of the local Gaussian approximation. Consequently, the **Euclidean covariance matrix** Σ must be defined as the inverse of the metric tensor:

$$\Sigma_\mu(x) = \alpha\, G^{-1}(\mu(x)) = \alpha\, P(\mu(x))$$

where $\alpha > 0$ is a scalar temperature parameter.
We do not need the full formalism of **Riemannian normal distributions**, we only enforce that the Euclidean covariance is aligned with the learned precision field.

Remark (Stability):
This definition resolves the tension between exploration and precision:

- **In Data Clusters:** The precision P is large. Therefore, the covariance $\Sigma \propto P$ is **large**.
  This allows the sampler to expand in coordinate space to cover the ***full extent of the dense data cluster***.
- **In The Void:** The precision P is small ($\approx \lambda I$). Therefore, the covariance $\Sigma \propto P$ is **small**.
  This constrains the sampler, preventing it from ***drifting into undefined regions*** where the decoder is untrained.

This formulation ensures that the encoder's uncertainty is strictly proportional to the manifold's local information density.

2.6 Relation to Existing Geometric VAE Frameworks

This work builds upon the geometric framework introduced by **Chadebec & Allassonnière (2020)**, specifically the construction of a Riemannian metric from local posterior covariances.
However, we introduce a fundamental inversion in the definition of the metric tensor to support trajectory modelling.

Definition 2.11 - **The Standard Geometric VAE Construction**

In the standard framework (e.g., RHVAE), the Riemannian metric $G_{std}(z)$ is identified directly with the local precision matrix (Fisher Information):

$$G_{std}(z) = \sum_{k=1}^{K} w_k(z)\Sigma_k^{-1} + \lambda I$$

Under this definition:

- **High-Density Regions:** Have high precision → **High Metric Cost** (G_{std} is large).
- **Geodesics:** Tend to curve *around* the high-density clusters to avoid the high cost, preferring the **void** where the metric is smaller (λI).
- **Prior:** The volume-based prior $p(z) \propto \sqrt{\det G_{std}(z)}$ correctly places mass on the data, but the geodesic structure repels dynamic trajectories from these regions.

Proposition 2.5 - **The Precision-Metric Inversion**

To enforce that latent trajectories flow **through** the data manifold rather than **around** it, the RLVAE inverts the relationship between ***information and metric cost***.
As defined in Section 2.2, we model the precision field $P(z)$ structurally identically to $G_{std}(z)$, but define our metric as its inverse:

$$G_{RLVAE}(z) \equiv P(z)^{-1} = G_{std}(z)^{-1}$$

Consequently:

- **High-Density Regions:** Have high precision → **Low Metric Cost** (G_{RLVAE} is small).
- **Geodesics:** Are attracted to the "valleys" of low cost, naturally following the data clusters.

Remark (The Precision-Weighted Prior):
Because we have inverted the metric, the standard Riemannian volume element $\sqrt{\det G_{RLVAE}}$ would assign high probability to the "void" (where G is large).
To recover a prior that concentrates on the data, we must define it with respect to the Precision Volume:

$$p(z) \propto \sqrt{\det G_{RLVAE}^{-1}(z)} = \sqrt{\det P(z)}$$

This formulation preserves the probabilistic desirable properties of the standard geometric VAE (prior mass on data) while inverting the dynamical properties (geodesics follow data). This "Precision-Constrained" geometry is essential for **generating coherent longitudinal interpolations**.

3 - Riemannian Posterior Construction (Stage C)

This section details the construction of the **variational posterior distribution** used in Stage C.
In contrast to classical VAEs, where the posterior is a parametric density (typically Gaussian), the RLVAE posterior is defined implicitly as the **pushforward** of a stochastic seed through a sequence of geometry-driven deterministic transformations.
The core principle is that the entire latent trajectory is generated from a single random pair (z_{in}, ρ_{in}), and all subsequent states are deterministic functions of this seed.
This ensures that inference remains probabilistic while enforcing strict geometric coherence and temporal consistency.

3.1 The Base Geometric Proposal

The entry point of the generative process is the **Base Proposal**, a probabilistic estimate of the initial state derived from the observed sequence.

Definition 3.1 - The Encoder Map

Let $\mathcal{X}^T$ be the space of observation sequences.
The **encoder** is a neural network $\mathcal{E}_\phi : \mathcal{X}^T \to \mathbb{R}^d$ that **maps** a sequence $\mathbf{x}_{0:T}$ to a latent mean vector $\mu = \mathcal{E}_\phi(\mathbf{x}_{0:T})$.

Definition 3.2 - The Riemannian Gaussian Proposal

To ensure the proposal respects the intrinsic geometry of the manifold learned in Stage B, we define the base distribution $q_{in}(z_{in}|\mathbf{x})$ as a **Gaussian in coordinates**, whose covariance is aligned with the **Riemannian metric**.
Consistent with the **Hypothesis 2.1** (**Precision-First**) and the **geometric covariance** definition (**Definition 2.9**), the covariance is proportional to the local information density:

$$q_{in}(z_{in}|\mathbf{x}) = \mathcal{N}\big(z_{in}; \mu, \Sigma_\phi(\mathbf{x})\big), \quad \text{with} \quad \Sigma_\phi(\mathbf{x}) = \alpha\, P(\mu),$$

where $P(\mu) = G^{-1}(\mu)$ is the precision field at the encoder mean and $\alpha > 0$ is a scalar temperature parameter.

Remark (Geometric Adaptivity):
This covariance structure ensures that the uncertainty of the proposal adapts to the manifold topology:

- **In High-Precision Regions** (Data Clusters): G_0^{-1} is large.
  The covariance Σ_ϕ is large, allowing the stochastic seed to explore the full extent of the cluster.
- **In Low-Precision Regions** (Voids): G_0^{-1} is small.
  The covariance Σ_ϕ is small, constraining the seed to remain close to the encoder's mean estimate μ.

This mechanism anchors the stochastic exploration to the informative regions of the latent space, providing a robust initialization for the subsequent Hamiltonian dynamics.

3.2 The Augmented Phase Space

To refine the initial proposal and align samples with the intrinsic structure of the manifold, we lift the latent variable z into the cotangent bundle $T^*\mathcal{M}$.
We augment the position state with an auxiliary momentum variable ρ, defining a Hamiltonian system that governs the geometric transport.

Definition 3.3 - Momentum Augmentation

For a given latent position z_{in}, we sample a **momentum vector** $\rho_{in} \in T^*\mathcal{M}$ from a local Riemannian Gaussian conditioned on the metric:

$$\pi(\rho_{in}|z_{in}) = \mathcal{N}\big(\rho_{in}; 0, G_0(z_{in})\big),$$

where $G_0(z)$ denotes the **base metric tensor** learned in Stage B, evaluated at the latent position z.

Remark:
This choice creates a mass matrix **equivalent** to the metric tensor $G_0(z)$.
Intuitively, **heavier regions** (high metric cost) require ***larger momentum to traverse***, ensuring the dynamics scale consistently with the local geometry.

Definition 3.4 - **The Precision-Driven Hamiltonian System**

We define the **Hamiltonian function** $H : T^*\mathcal{M} \to \mathbb{R}$ as the sum of potential and kinetic energies:

$$H(z, \rho) = U(z) + K(z, \rho)$$

- **Kinetic Energy:** Defined by the Riemannian norm of the momentum vector:

$$K(z, \rho) = \frac{1}{2}\rho^\top G_0(z)^{-1}\rho = \frac{1}{2}\rho^\top P(z)\rho$$

  Although $P(z)$ represents the information density field, in the Hamiltonian formulation it strictly plays the role of the **inverse mass matrix**, ensuring consistency between the stochastic momentum initialization and the deterministic dynamics.

- **Potential Energy:** Defined as the negative log-density of the geometric prior:

$$U(z) = -\log p_{prior}(z) = -\frac{1}{2}\log\det G_0^{-1}(z)$$

  This choice corresponds to the **Precision-Weighted Prior** introduced in Part V, ensuring that the Hamiltonian dynamics naturally pull samples towards regions of high information density.
  Unlike classical RHMC where the potential reflects a Euclidean prior, here the potential is explicitly driven by the learned precision field $P(z)$.
  Note that this convention ensures that **high precision regions** correspond to **low potential energy**, making them attractors of the **Hamiltonian dynamics**.

Proposition 3.1 - **The Refinement Force**

The **gradient of the potential energy**, which drives the momentum update, is given by:

$$\nabla_z U(z) = -\nabla_z \left(\frac{1}{2}\log\det P(z)\right)$$

Interpretation:

The Hamiltonian dynamics are driven explicitly by the gradient of the information density.
The force field $\nabla_z U$ accelerates the particle towards regions of higher precision (local maxima of $\det P$).
Consequently, the system does not merely sample the encoder's uncertainty: it actively **refines the sample**.
This refinement is deterministic conditioned on the initial seed (z_{in}, ρ_{in}), explaining why RLVAE posterior samples concentrate on the intrinsic manifold structure even when the initial encoder proposal is diffuse.

3.3 The Geometric Projector (RHMC Transport)

The transition from the proposal distribution to the valid initial trajectory state is mediated by a deterministic operator Φ.
We construct this operator by integrating the Hamiltonian system defined in **Section 3.2** over a fixed pseudo-time window.

Definition 3.5 - **The Geometric Refinement Operator**

Let $(z_{in}, \rho_{in}) \in T^*\mathcal{M}$ be the initial phase space configuration drawn from the encoder.
We define the **Geometric Refinement Operator** $\Phi : T^*\mathcal{M} \to T^*\mathcal{M}$ as the time-evolution map of the Hamiltonian system:

$$(z_0, \rho_0) = \Phi(z_{in}, \rho_{in})$$

Remark:

Crucially, Φ is **not** used here as a **Markov transition operator** targeting the prior distribution (as in standard MCMC).
Instead, it acts as a **finite-time geometric transport** that reshapes the support of the encoder proposal such that it aligns with the learned manifold geometry, without destroying the semantic structure of the initial guess.

Algorithm 3.1 - Riemannian Leapfrog Integration

In practice, Φ is approximated by L steps of a **Generalized Leapfrog Integrator** with step size ϵ.
Since the kinetic energy $K(z,\rho)$ depends on position via the metric, the equations of motion include metric gradients.
A single **integration step** $(z,\rho) \to (z',\rho')$ proceeds as:

1. **Momentum Half-Step:** Update momentum using the gradient of the total energy:

$$\rho_{1/2} \leftarrow \rho - \frac{\epsilon}{2}\left(\nabla_z U(z) + \nabla_z K(z,\rho)\right)$$

   (Note: $\nabla_z K$ includes the partial derivatives of the metric tensor $G_0(z)$).

2. **Position Full-Step:** Update position using the velocity derived from the precision matrix:

$$z' \leftarrow z + \epsilon\, G_0^{-1}(z)\, \rho_{1/2}$$

3. **Momentum Half-Step:** Update momentum using gradients at the new position:

$$\rho' \leftarrow \rho_{1/2} - \frac{\epsilon}{2}\left(\nabla_z U(z') + \nabla_z K(z',\rho_{1/2})\right)$$

Proposition 3.2 - Properties of the Projector

The operator Φ satisfies the following properties essential for the variational bound:

- **Determinism:** Given the input pair (z_{in},ρ_{in}), the output (z_0,ρ_0) is **unique** and **deterministic**.
- **Diffeomorphism:** The map is **differentiable** with respect to its inputs, allowing gradients to **backpropagate** from the reconstruction loss to the encoder.
- **Quasi-Symplectic Structure:** While theoretical Hamiltonian flow preserves volume ($|\det J| = 1$), numerical integration introduces small errors.
  The volume distortion $\Delta_{vol} = \log|\det J_\Phi|$ is non-zero but bounded, and **must be explicitly included** in the variational objective to maintain the correctness of the change-of-variables formulation (see Section 5.5).

Proof:
Let the operator Φ be the composition of L steps of the generalized leapfrog integrator defined in Algorithm 3.1.
We analyze a single integration step $S : (z,\rho) \mapsto (z',\rho')$, which consists of three sub-steps $S = S_3 \circ S_2 \circ S_1$.

1. **Determinism:**
   The update equations in Algorithm 3.1 depend only on the state (z,ρ), the metric tensor $G_0(z)$ (and its derivatives), and the potential $U(z)$.
   By construction (Part II), G_0 and U are **deterministic** and smooth functions of z.
   The integrator does not introduce any additional randomness: once the seed (z_{in},ρ_{in}) is fixed, all subsequent updates are fixed.
   Therefore, the output $(z_0,\rho_0) = \Phi(z_{in},\rho_{in})$ is **unique**, and Φ is **deterministic**.
2. **Local Diffeomorphism:**
   Global invertibility is ***not required*** for the variational formulation, only **local differentiability** and a **well-defined Jacobian** almost everywhere.
   We examine the Jacobian of each sub-step in the augmented phase space (z,ρ):
   - **Sub-step 1** (momentum half-step): $\rho_{1/2} = \rho - \frac{\epsilon}{2}\mathcal{F}(z,\rho)$.
     This is an affine map in ρ whose Jacobian is a smooth perturbation of the identity.
     For sufficiently small ϵ, the Jacobian block with respect to ρ remains non-singular, so this map is locally invertible and differentiable.
   - **Sub-step 2** (position full-step): $z' = z + \epsilon\, G_0^{-1}(z)\, \rho_{1/2}$.
     By the spectral bounds established in Part II, $G_0^{-1}(z)$ is smooth with bounded eigenvalues on the region explored by the sampler.
     The Jacobian with respect to z,

$$J_{S_2}(z,\rho_{1/2}) = I + \epsilon \frac{\partial}{\partial z}\left(G_0^{-1}(z)\rho_{1/2}\right),$$

     is a **smooth perturbation** of the identity.
     For step sizes ϵ small enough (as enforced in practice for stability), this matrix is non-singular via a Neumann series

argument, and the map is **locally invertible**.

- **Sub-step 3 (momentum half-step):** Symmetric to Sub-step 1, with the same properties.

Since Φ is a finite composition of such locally invertible smooth maps, it is itself a local diffeomorphism.
This guarantees that gradients from the reconstruction loss can backpropagate through Φ to the encoder parameters.

3. Quasi-Symplectic Structure and Volume Distortion:
For a continuous Hamiltonian flow on $T^*\mathcal{M}$, Liouville's theorem guarantees exact preservation of phase-space volume ($|\det J| = 1$).
In Euclidean HMC with a constant mass matrix, the standard leapfrog integrator is symplectic and therefore exactly volume-preserving at the discrete level.
In our Riemannian setting, the position update in **Sub-step 2** depends non-linearly on z through the metric $G_0(z)$.
For the explicit generalized leapfrog scheme used in our implementation, the Jacobian determinant of **Sub-step 2** is:

$$\det J_{S_2} = \det\left(\frac{\partial z'}{\partial z}\right) = \det\left(I + \epsilon \frac{\partial}{\partial z}\left(G_0^{-1}(z)\rho_{1/2}\right)\right),$$

which is, in general, not equal to 1 when the ***metric varies spatially***.
As a result, the total Jacobian of the full map satisfies:

$$\log|\det J_\Phi| = \sum_{l=1}^{L} \log|\det J_S^{(l)}| \neq 0$$

This deviation corresponds to a controlled contraction or expansion of phase-space volume induced by the discrete integrator.
The deviation from unit determinant is typically $O(\epsilon)$, and empirically remains bounded due to the spectral constraints on G_0.
To maintain a valid change-of-variables formula in the ELBO (Section 5.5), we explicitly compute this quantity in practice and incorporate it as the volume-distortion correction term $\Delta_{vol} = \log|\det J_\Phi|$ in the KL.

Interpretation (The "Magnet" Effect):
The operator Φ acts as a Finite-Time Projector:

- The Potential Force $-\nabla_z U$ pulls the latent code towards the **centroids** where ***precision is high***.
- The Kinetic Term allows the particle to traverse the **void** (low precision) rapidly without getting trapped, utilizing the momentum acquired from the encoder's uncertainty.
- Control**:** The integration horizon $L\epsilon$ controls the sharpness of the projection:
  - as $L\epsilon \to 0$, Φ approaches the identity (trusting the encoder).
  - as $L\epsilon$ increases, z_0 converges strictly toward the manifold core defined by the precision field $P(z)$.

3.4 The Longitudinal Flow

Once the initial latent state z_0 has been geometrically refined, the full longitudinal trajectory $\mathbf{z}_{1:T}$ is generated deterministically by a sequence of invertible transformations.
These flows model the continuous **temporal evolution of the system** (e.g., biological growth or disease progression) as a transport process on the manifold.

Definition 3.6 - The Flow Sequence

Let $\mathcal{F} = \{f_t\}_{t=1}^T$ be a sequence of diffeomorphisms $f_t : \mathcal{M} \to \mathcal{M}$, implemented in Euclidean coordinates via neural networks.
The latent state at time t is defined recursively:

$$z_t = f_t(z_{t-1}), \quad \forall t \in \{1, \ldots, T\}$$

We denote the Composite Transport Map $F : \mathcal{M} \to \mathcal{M}$ as the function that maps the initial state z_0 directly to the final state z_T :

$$z_T = F(z_0) = f_T \circ f_{T-1} \circ \cdots \circ f_1(z_0)$$

Remark:

While the flow transformations operate on Euclidean coordinates, their probabilistic evaluation and geometric interpretation are performed strictly with respect to the transported Riemannian metric G_t, ensuring consistency between the learned geometry and the temporal dynamics.

Proposition 3.3 - Jacobian Accumulation

Since the flows are **invertible** and **differentiable**, the probability density of the trajectory transforms according to the ***change of variables formula***.
The log-determinant of the Jacobian of the composite map F is the sum of the log-determinants of the individual steps:

$$\log|\det J_F(z_0)| = \sum_{t=1}^{T} \log\left|\det \frac{\partial f_t}{\partial z_{t-1}}(z_{t-1})\right|$$

Remark (Expansion):
The sign and magnitude of this Jacobian determinant are critical.

- If $\log|\det J_F| > 0$, the flow **expands** the latent coordinate volume.
- If $\log|\det J_F| < 0$, the flow **contracts** the volume.

As derived in Part V, since the Precision-Weighted Prior density increases as the **geometric volume element contracts** in data regions, the flow is mathematically incentivized to stretch coordinate volume ($\det J \to \infty$) to maximize the prior probability.
This creates an intrinsic **Expansion Drive** that necessitates the explicit regularization derived in **Section 5.6**.

Dynamic Geometry Transport

Crucially, the flow does not just move the point z; it transports the manifold structure itself.
The metric tensor at time t, denoted G_t, is obtained via the pushforward of the base metric G_0 by the composite flow.
Explicitly, the metric transforms as a covariant 2-tensor:

$$G_t(z_t) = (f_{1:t})_* G_0(z_0) = \left(J_F(z_0)^{-1}\right)^\top G_0(z_0)\, J_F(z_0)^{-1},$$

where $J_F(z_0) = \frac{\partial z_t}{\partial z_0} = \frac{\partial F(z_0)}{\partial z_0}$ is the Jacobian of the cumulative transport.

This ensures that the notion of **distance** and **precision** evolves coherently with the data.
A region that is small and dense at $t = 0$ might expand into a large, complex region at $t = T$, and the metric deforms (via the Jacobian inverse) to track this growth, **preserving** the underlying **topological structure**.

3.5 The Posterior as a Pushforward Measure

We now summarize the complete generative process for the variational posterior.
The distribution $q_\phi(\mathbf{z}_{0:T}|\mathbf{x})$ is constructed by pushing a simple stochastic seed through a sequence of geometry-driven deterministic maps.

Definition 3.7 - The Sampling Pipeline

The generation of a single latent trajectory sample proceeds in three distinct phases:

1. **Stochastic Initialization (The Seed):**
   The encoder network maps the observation $\mathbf{x}$ to a proposal distribution on the cotangent bundle $T^*\mathcal{M}$.
   We sample an initial position and momentum:

$$z_{in} \sim q_{in}(z|\mathbf{x}) = \mathcal{N}\left(\mu, \alpha G_0^{-1}(\mu)\right), \quad \rho_{in} \sim \mathcal{N}\left(0, G_0(z_{in})\right)$$

2. **Geometric Refinement (The Projector):**
   The seed is projected onto the high-precision manifold shell via the Hamiltonian operator:

$$(z_0, \rho_0) = \Phi(z_{in}, \rho_{in})$$

3. **Temporal Transport (The Flow):**
   The refined state is transported through time to generate the full trajectory:

$$z_t = f_t(z_{t-1}), \quad \forall t \in \{1, \ldots, T\}$$

Proposition 3.4 - The Trajectory Posterior

Since the transformation from (z_{in}, ρ_{in}) to $\mathbf{z}_{0:T}$ is deterministic, the variational posterior is a ***singular distribution supported*** on a d-dimensional immersed submanifold of the trajectory space $\mathcal{M}^{T+1}$, parameterized by the initial seed via the smooth map $F \circ \Phi$.
It can be expressed formally using Dirac measures:

$$q_\phi(\mathbf{z}_{0:T}|\mathbf{x}) = q_{in}(z_{in}|\mathbf{x})\pi(\rho_{in}|z_{in}) \cdot \delta\big(z_0 - \Phi_z(z_{in}, \rho_{in})\big) \cdot \prod_{t=1}^{T} \delta\big(z_t - f_t(z_{t-1})\big)$$

Equivalently, in pushforward form:

$$q_\phi(\mathbf{z}_{0:T}|\mathbf{x}) = (F \circ \Phi)_\# \left[q_{in}(z_{in}|\mathbf{x})\, \pi(\rho_{in}|z_{in})\right]$$

Interpretation:
This formulation highlights the fundamental shift from classical VAEs:

- **Probabilistic Source:** All stochasticity is confined to the initial seed (z_{in}, ρ_{in}). The model captures epistemic uncertainty *before* interaction with the manifold geometry.
- **Geometric Determinism:** Once seeded, the trajectory is fully constrained by the learned geometry (G_0) and dynamics (f_t).
- **Dimension Reduction:** Although the seed space is $2d$-dimensional (position + momentum), only the position component determines the observable trajectory, while momentum only affects the geometric refinement path, resulting in a d-dimensional posterior manifold.

This structure allows us to evaluate the variational objective (ELBO) by simply integrating over the initial seed and applying the change-of-variables formula along the deterministic path, as derived in **Part V**.

4 - Geometrically Induced Prior and Dynamic Regularisation

This section formalises the prior distribution used in the RLVAE framework.
Unlike classical VAEs that assume a fixed isotropic prior $p(z) = \mathcal{N}(0, I)$, or static Riemannian VAEs that use a fixed geometric prior $p(z) \propto \sqrt{\det G(z)}$, our model requires a prior that is:

- **Precision-Weighted:** Favoring regions of high structural information rather than high volume.
- **Time-Dependent:** Evolving coherently with the latent flow dynamics.
- **Trajectory-Structured:** Regularizing the entire path $\mathbf{z}_{0:T}$ rather than independent states.

We first establish the conceptual role of this prior, then rigorously derive its density on the base manifold (Stage B) and its transport through time (Stage C).

4.1 Conceptual Role of the Prior in RLVAE

In a variational framework, the prior serves as the **fundamental regularization mechanism**, encoding structural assumptions about the latent space independently of observed data.

Within the RLVAE, this role is intrinsically geometric and longitudinal, and fulfils two complementary objectives:

1. **Structural Guidance:** enforcing confinement of latent states to geometrically credible regions of the manifold.
2. **Dynamical Constraint:** shaping the evolution of trajectories through the induced energy landscape.

Remark 4.1 - Prior as a Geometric Constraint

In **Euclidean** space, the prior $p(z)$ acts as a **centripetal force**, encouraging latent representations to remain **compact** (typically around the origin via an isotropic Gaussian).
On a **Riemannian** manifold $\mathcal{M}$, the prior must act as a **Manifold Constraint Force**, restricting latent trajectories to the subspace of geometrically valid configurations.

- **Standard Geometric VAEs** (RHVAE) employ volume-based priors of the form: $p_{vol}(z) \propto \sqrt{\det G(z)}$, which concentrate probability mass in regions of large metric volume.
- The RLVAE adopts a **Precision-First** paradigm: the prior must attract samples toward regions of **high structural confidence**, characterised by high precision (G^{-1} large) and low metric cost (G small), corresponding to dense data clusters identified during Stage B.

Hypothesis 4.1 - Dynamic Geometry Principle

The latent trajectory evolves according to a sequence of invertible flows:

$$z_t = f_t(z_{t-1})$$

Since these flows deform the latent coordinate system, the geometry itself must be transported accordingly to preserve coherence between structure and dynamics.
Therefore, the prior **cannot be static**.
It must be defined as the **pushforward of the base geometric prior through the flow dynamics**, yielding a time-dependent distribution $p_t(z)$ that evaluates the geometric plausibility of each state within its local coordinate system.
This ensures that regularisation adapts to the temporal deformation of the manifold and faithfully penalises deviations from geometrically valid evolution paths.

4.2 The Base Riemannian Structure (Measure-Theoretic Foundation)

This section formalises the static geometric object from which the entire dynamic prior is constructed. We explicitly distinguish between:

- the **extrinsic coordinate representation** used by neural networks and flows,
- the **intrinsic geometric structure** governing distances and volume on the manifold.

This distinction is essential for understanding why the RLVAE prior induces an intrinsic incentive for coordinate expansion.

Definition 4.1 - The Base Riemannian Manifold

Let $\mathcal{Z} \cong \mathbb{R}^d$ denote the global coordinate chart of the latent space.
We define the **Base Latent Manifold** as the Riemannian manifold:

$$\mathcal{M}_0 = (\mathcal{Z}, G_0),$$

where the metric tensor field is constructed from the precision field learned in Stage B.
Consistent with the **Precision-First Hypothesis** (Hypothesis 2.1), the metric tensor is defined as:

$$G_0(z) = P_0(z)^{-1} = \left(\lambda I_d + \sum_{k=1}^{K} w_k(z)\Lambda_k\right)^{-1},$$

where:

- $P_0(z)$ encodes local information density,
- $G_0(z)$ encodes geometric cost (local stretch/compression),
- $z \in \mathcal{Z}$.

Thus, geometry is not identified with Euclidean coordinates, but with the **local deformation** induced by the **learned precision field**.

Definition 4.2 — Coordinate Measure vs Riemannian Measure

The neural architectures operating in the latent space (encoder and flows) implicitly treat $\mathcal{Z}$ as a Euclidean vector space, inducing the **coordinate reference measure**:

$$dz \quad \text{(Lebesgue measure)}$$

However, the manifold $(\mathcal{M}_0, G_0)$ admits an intrinsic volume form derived from the metric tensor.
The associated **Riemannian Volume Measure** is defined as:

$$d\mathrm{Vol}_{G_0}(z) \;=\; \sqrt{\det G_0(z)}\, dz$$

Using the Precision-First convention, this becomes:

$$d\mathrm{Vol}_{G_0}(z) = \frac{1}{\sqrt{\det P_0(z)}}\, dz$$

This establishes a **fundamental asymmetry** between **coordinate space** and **intrinsic geometry**.

Remark (Coordinate vs Intrinsic Geometry):

This distinction encodes the core geometric interpretation of Stage B:

- **High-Precision Regions ($P_0(z)$ large)**

  $\rightarrow G_0(z)$ **small** $\rightarrow \det G_0(z)$ small $\rightarrow$ intrinsic geometric volume is **compressed!**

  A large coordinate neighbourhood corresponds to a small intrinsic manifold volume.

- **Low-Precision Regions ($P_0(z) \approx \lambda I$)**

  $\rightarrow G_0(z)$ **large** $\rightarrow \det G_0(z)$ large $\rightarrow$ intrinsic volume is **expanded!**

  A small coordinate neighbourhood corresponds to a large intrinsic manifold volume.

Hence, Euclidean coordinates do not faithfully represent geometric density.
The metric introduces a deformation field that **reshapes space** according to information content.

Definition 4.3 - Geometric Density Field

We define the **intrinsic density field** associated with the manifold as:

$$\nu_0(z) := \sqrt{\det G_0(z)} = \frac{1}{\sqrt{\det P_0(z)}}$$

This quantity measures the **relative** **expansion** or **compression** of the manifold with respect to Euclidean coordinates.
It quantifies how many intrinsic ***units of volume*** are represented by one unit of coordinate space.

Proposition 4.1 - Volume Distortion Induced by the Metric

For any measurable region $A \subset \mathcal{Z}$, the **intrinsic volume** satisfies:

$$\mathrm{Vol}_{G_0}(A) = \int_A \sqrt{\det G_0(z)}\, dz$$

Thus, coordinate regions containing **high-precision centroids** contribute disproportionately **little** intrinsic volume, while **void regions** contribute disproportionately **large** intrinsic volume.

Remark (Origin of the Expansion Incentive):

If a probability density $p(z)$ is defined with respect to the coordinate measure dz, but **regularisation** is performed relative to the **Riemannian volume**, then **maximising likelihood** encourages **expansion** in coordinate space because **intrinsic mass** density concentrates in compressed regions.
This mismatch creates a natural tension:

Object	Measure
Flows operate in	Coordinate space (dz)
Geometry measured in	Intrinsic space ($d\mathrm{Vol}_{G_0}$)

This tension is the mathematical origin of the **Expansion Drive** formalised in Section 4.4.

Interpretation Summary

The base manifold introduces a non-uniform relationship between: **Coordinate volume** + **Geometric meaning** + **Probabilistic regularisation**
The latent space is therefore: **Not** **flat** **in** ***intrinsic geometry*** + **Not** **isotropic** **in** ***information density*** + **Not** **symmetric** **under** ***Euclidean translations***
This prepares the foundation for defining a prior that aligns probabilistic mass with high-precision structures while preserving coordinate flexibility.

4.3 The Precision-Weighted Base Prior

Having established the existence of a Riemannian volume structure on the base manifold $\mathcal{M}_0 = (\mathcal{Z}, G_0)$, we now define the probabilistic prior distribution that governs the regularisation of latent states.

The critical departure from classical geometric VAEs lies in the inversion of the volume principle: instead of distributing probability mass proportionally to intrinsic volume, we concentrate mass in regions of high geometric compression, interpreted as high structural confidence.

Definition 4.4 - **The Precision-Weighted Prior Density**

Let $\mathcal{M}_0 = (\mathcal{Z}, G_0)$ be the base Riemannian manifold defined in **Section 4.2**.
We define the **Precision-Weighted Base Prior** as a probability measure p_0 on $\mathcal{Z}$, absolutely continuous with respect to the Lebesgue measure dz, with density:

$$p_0(z) = \frac{1}{Z_0}\sqrt{\det G_0^{-1}(z)} = \frac{1}{Z_0}\sqrt{\det P_0(z)},$$

where the normalising constant is:

$$Z_0 = \int_{\mathcal{Z}} \sqrt{\det P_0(z)}\, dz < \infty$$

Remark (Interpretation as Dual to Riemannian Volume):
Recall that the Riemannian volume density is

$$\nu_0(z) = \sqrt{\det G_0(z)} = \frac{1}{\sqrt{\det P_0(z)}}$$

Thus:

$$p_0(z) \propto \frac{1}{\nu_0(z)}$$

The prior therefore assigns **maximum probability density to geometrically compressed regions**, which correspond to high information density in the learned latent manifold.
This is the mathematical embodiment of the **Precision-First paradigm**!

Proposition 4.2 - **Structural Concentration Property**

Let c_k denote a centroid of the precision field (Stage B), and consider a **local neighbourhood** U_k centred at c_k. Then:

$$z \in U_k \;\Rightarrow\; \forall z' \text{ in void regions}, \quad p_0(z) \gg p_0(z')$$

Hence, the prior deliberately biases the latent space toward the **empirical support of the data manifold**.

Remark (Contrast with Classical Riemannian Priors):
In standard geometric VAEs (e.g. RHVAE), the prior is defined as:

$$p_{\mathrm{vol}}(z) \propto \sqrt{\det G_0(z)}$$

This allocates mass proportionally to intrinsic volume, meaning: **expanded regions** receive **higher probability** + **compressed regions** receive **lower probability**.
Under the Precision-First metric ($G_0 = P_0^{-1}$), this would force probability mass into **geometrically large voids** — the exact opposite of the modelling objective.
Thus, the RLVAE prior deliberately inverts this behaviour:

$$p_0(z) \propto \frac{1}{\sqrt{\det G_0(z)}}$$

Definition 4.5 - **Prior Energy (Geometric Potential)**

The negative log-prior defines the **geometric energy function**:

$$U_0(z) = -\log p_0(z) = -\frac{1}{2}\log\det P_0(z) + \log Z_0$$

This quantity has two simultaneous conceptual roles:

1. **Regularisation term** in the ELBO.
2. **Potential energy** guiding RHMC dynamics

This dual-use ensures that sampling dynamics and variational regularization are governed by the **same geometric principle**.

Proposition 4.3 - Attraction Toward Structural Manifold

The gradient of the prior energy is given by:

$$\nabla_z U_0(z) = -\nabla_z \left(\frac{1}{2} \log \det P_0(z) \right),$$

which produces a **force field** proportional to the **gradient of information density**.
Consequently:

- Latent states are actively **pulled toward** **high-precision centroids.**
- Trajectories are **discouraged** from drifting into **low-confidence regions.**

This enforces structural coherence at both geometric and probabilistic levels.

Remark (Origin of the Expansion Drive):
Although the prior concentrates mass at centroids, when combined with ***change-of-variables*** under Euclidean flows, the model becomes incentivized to:

- **Increase** coordinate volumes
- **Stretch** the latent space

to counterbalance high prior density.
This creates the intrinsic **Expansion Drive**, whose corrective regularisation is introduced in Section 5.6.

Interpretative Summary

Feature	Effect
$\sqrt{\det P_0(z)}$	Concentrates prior mass in **data-dense** region
$U_0(z)$	Energy guiding RHMC refinement
∇U_0	Force pulling **toward centroids**
Coordinate–geometry mismatch	Produces expansion incentive

This prior therefore simultaneously:

- **Anchors latent states** to meaningful **structure**.
- **Penalises** **geometric** drift.
- Drives controlled deformation of **coordinate space**.

4.4 Transported Prior and Dynamic Geometry Regularisation

The base prior defined in **Section 4.3** governs the geometry at the initial latent time.
However, latent states do not evolve on a static manifold: both the coordinates and the geometry are advected by the longitudinal flow.
We now define the **time-dependent prior** induced by transporting the base geometric measure along the deterministic trajectory map.

Definition 4.6 **- The Transported Manifold**

Let $F_t = f_t \circ \cdots \circ f_1$ denote the composite flow from time 0 to time t.
The latent space at time t is the Riemannian manifold

$$\mathcal{M}_t = (\mathcal{Z}, G_t),$$

where the metric G_t is the **pushforward** of the base metric G_0:

$$G_t(z_t) = (F_t)_* G_0(z_0) = \left(J_{F_t}(z_0)^{-1}\right)^\top G_0(z_0)\, J_{F_t}(z_0)^{-1}$$

Consequently, the **precision field** $P_t = G_t^{-1}$ transforms contravariantly:

$$P_t(z_t) = J_{F_t}(z_0)\, P_0(z_0)\, J_{F_t}(z_0)^\top$$

Definition 4.7 **- Time-Dependent Geometric Prior**

Consistent with the Precision-First paradigm, the prior at time t is defined as the density proportional to the square root of the transported precision determinant:

$$p_t(z_t) = \frac{1}{\mathcal{Z}_t} \sqrt{\det P_t(z_t)},$$

where $\mathcal{Z}_t$ is a normalizing constant.
This defines a **probability density** with respect to Lebesgue measure that is **uniform** **with respect to the** **transported precision-induced measure**, not a mass-preserving pushforward of p_0.

Lemma 4.1 **- No Mass Transport**

The dynamic prior p_t is ***not*** the pushforward of the base prior p_0 as a **probability measure**.
Standard probability transport conserves mass:

$$\int_A p_0(z)\, dz = \int_{F_t(A)} p_t(z_t)\, dz_t,$$

implying density decreases under expansion.
In contrast, the RLVAE prior is recomputed from the **transported metric** at each time step:

$$p_t(z_t) \propto \sqrt{\det P_t(z_t)}$$

We therefore transport **geometry**, not probability mass, this distinction is the foundation of **precision-driven regularisation**.

Proposition 4.4 **- Explicit Density and Expansion**

The time-dependent prior satisfies:

$$p_t(z_t) \propto p_0(z_0)\, |\det J_{F_t}(z_0)|$$

Proof:

From Definition 4.6:

$$\det P_t(z_t) = \det(J P_0 J^\top) = (\det J)^2 \, \det P_0(z_0)$$

Hence,

$$p_t(z_t) \propto \sqrt{\det P_t(z_t)} = |\det J_{F_t}| \sqrt{\det P_0(z_0)} \propto p_0(z_0)\, |\det J_{F_t}(z_0)|$$

Remark (The Expansion Anomaly):
This exposes a fundamental divergence:

- **Standard Probability (Posterior Pushforward):**

$$q_t(z_t) = q_0(z_0)\,|\det J_{F_t}|^{-1}$$

so **expansion decreases density**.

- **Geometric Precision (Prior Construction):**

$$p_t(z_t) = p_0(z_0)\,|\det J_{F_t}|$$

so **expansion increases density**.

This creates the **Expansion Drive**: the model can artificially increase prior likelihood by expanding coordinate volume.

Proposition 4.5 - Prior Contribution to Regularisation

The negative log prior decomposes as:

$$\log p_t(z_t) = \underbrace{U_0(z_0)}_{\text{Base Precision Attraction}} - \underbrace{\log|\det J_{F_t}(z_0)|}_{\text{Prior-Induced Expansion}},$$

where $U_0(z_0) = -\log p_0(z_0)$.
Combined with the posterior entropy term

$$\log q_t = \log q_0 - \log|\det J_{F_t}|,$$

the total Jacobian contribution to the KL divergence becomes:

$$2\log|\det J_{F_t}|,$$

which grows **unbounded under expansion**, formally explaining the instability of the unregularized objective.

> Had we instead defined the prior as a **probability pushforward** of p_0, the Jacobian terms would **cancel** in the KL divergence, eliminating the expansion drive and yielding numerical stability.
> However, such a prior would be **blind to the manifold geometry** and would fail to constrain trajectories to structurally meaningful regions.
> The **Expansion Drive** is therefore the direct mathematical signature of **geometry-aware regularization**.

4.5 Consequences of the Precision-Based Prior and Necessity of Regularisation

The dynamic prior constructed in Section 4.4 is geometrically well-founded and consistent with the Precision-First paradigm.
However, its interaction with the longitudinal flow introduces a **structural instability that must be explicitly addressed**.

Definition 4.8 - Geometric Regularisation Energy

The regularisation contribution of the prior at time t is defined by:

$$\mathcal{R}_t(z_t) = -\log p_t(z_t)$$

Using the result of Proposition 4.5, this can be decomposed as:

$$\mathcal{R}_t(z_t) = U_0(z_0) - \log|\det J_{F_t}(z_0)|$$

The first term enforces **attraction towards the core** of the learned manifold, while the second introduces an incentive for the longitudinal flow to **expand volume**.

Lemma 4.2 - Unboundedness Under Volume Expansion

The geometric regularisation term $\mathcal{R}_t$ is **unbounded** below under **unconstrained expansion of the flow volume**.

> **Proof:**
> If $|\det J_{F_t}(z_0)| \to \infty$, then $-\log p_t(z_t) \to -\infty$.
> Hence, the **optimisation problem** admits **degenerate solutions** characterized by arbitrarily large Jacobian determinants!

Proposition 4.6 - Expansion Instability

In absence of **explicit constraints** or counter-regularisation, the precision-based prior induces a **pathological optimum** where the longitudinal flow maximises volume expansion.
Consequently, the variational objective becomes dominated by **geometric deformation** rather than reconstruction fidelity or semantic consistency.
This instability is a direct consequence of **enforcing geometry awareness**: the model is rewarded for increasing coordinate volume in high-precision regions, thereby artificially inflating the **prior probability**.

Remark (Geometry vs Stability):
This expansion tendency is **not an implementation artefact** but a mathematical property of **precision-based geometry!**
It stems from the dissociation between intrinsic geometric density and coordinate probability mass.
Classical VAEs conceal this behaviour by adopting **Euclidean Gaussian priors**, which implicitly enforce contraction.
The RLVAE instead makes this interaction explicit, ***requiring a complementary force to ensure equilibrium***.

→ **The resolution of this instability** does not require abandoning the precision-based prior.
Instead, it motivates the introduction of an **explicit counter-term** balancing the **Expansion Drive**.
This leads to a geometrically normalized variational objective, formally derived in **the next part!**

5 - Full ELBO Derivation under Deterministic Geometric Dynamics

This section provides the complete derivation of the **variational objective** optimized by the RLVAE in Stage C.
We begin by establishing the general **E**vidence **L**ower **BO**und (**ELBO**) for longitudinal data sequences.
We then demonstrate how the imposition of deterministic Riemannian dynamics collapses the path-wise KL divergence into a boundary term, and finally expand the objective to account for the specific precision-weighted geometry and numerical integration corrections.

5.1 General Longitudinal ELBO

Let $\mathcal{X}$ denote the observation space (e.g., image space) and $\mathcal{M} \cong \mathbb{R}^d$ denote the latent manifold.
We consider a longitudinal dataset consisting of a sequence of observations $\mathbf{x}_{0:T} = \{x_0, x_1, \dots, x_T\}$, where each $x_t \in \mathcal{X}$.
We assume these observations are generated by a corresponding sequence of latent variables $\mathbf{z}_{0:T} = \{z_0, z_1, \dots, z_T\}$, where each $z_t \in \mathcal{M}$.

5.1.1 The Generative Model

Definition 5.1 - Longitudinal Generative Model

We define the **Joint Distribution** of the observations and latent trajectory by:

$$p_\theta(\mathbf{x}_{0:T}, \mathbf{z}_{0:T}) = p(\mathbf{z}_{0:T}) \prod_{t=0}^{T} p_\theta(x_t \mid z_t),$$

where:

- $p(\mathbf{z}_{0:T})$ is the **prior distribution** over the full latent trajectory,
- $p_\theta(x_t \mid z_t)$ is the **decoder likelihood** (observation model) parameterised by θ.

Remark (Conditional Independence):
The factorisation above corresponds to the standard assumption that observations at time t are **conditionally independent** given the latent state z_t.
In the context of **disease progression imaging**, this hypothesis assumes that each image represents an **independent observation** of the current physiological state of the patient. Temporal correlations observed across scans are therefore attributed to the **evolution of the underlying pathology**, not to **memory effects in the imaging process itself**.
This aligns with the clinical view that sequential scans reflect successive snapshots of a continuously evolving biological process.

MAYBE WORK ON THE HYPOTHESIS MORE !!!!

5.1.2 The Variational Lower Bound

To learn θ, our goal is to maximize the marginal log-likelihood of the data $\log p_\theta(\mathbf{x}_{0:T})$.

Since the **marginalization** over $\mathbf{z}_{0:T}$ is **intractable**, we introduce a **variational posterior**!

Definition 5.2 - Variational Trajectory Posterior

Let

$$q_\phi(\mathbf{z}_{0:T} \mid \mathbf{x}_{0:T})$$

be a **variational distribution over latent trajectories**, parameterised by ϕ, used to approximate the **true posterior**:

$$p_\theta(\mathbf{z}_{0:T} \mid \mathbf{x}_{0:T})$$

Proposition 5.1 - Evidence Lower BOund (ELBO)

For any choice of $q_\phi(\mathbf{z}_{0:T} \mid \mathbf{x}_{0:T})$, the following inequality holds:

$$\log p_\theta(\mathbf{x}_{0:T}) \;\geq\; \mathcal{L}(\theta, \phi; \mathbf{x}_{0:T}),$$

where

$$\mathcal{L}(\theta, \phi; \mathbf{x}_{0:T}) = \int_{\mathcal{M}^{T+1}} q_\phi(\mathbf{z}_{0:T} \mid \mathbf{x}_{0:T}) \log \frac{p_\theta(\mathbf{x}_{0:T}, \mathbf{z}_{0:T})}{q_\phi(\mathbf{z}_{0:T} \mid \mathbf{x}_{0:T})} \, d\mathbf{z}_{0:T}$$

Proof:
Starting from the marginal likelihood:

$$\log p_\theta(\mathbf{x}_{0:T}) = \log \int p_\theta(\mathbf{x}_{0:T}, \mathbf{z}_{0:T}) \, d\mathbf{z}_{0:T},$$

we multiply and divide by $q_\phi(\mathbf{z}_{0:T} \mid \mathbf{x}_{0:T})$ and apply Jensen's inequality:

$$\begin{aligned}
\log p_\theta(\mathbf{x}_{0:T}) &= \log \int q_\phi(\mathbf{z}_{0:T} \mid \mathbf{x}_{0:T}) \frac{p_\theta(\mathbf{x}_{0:T}, \mathbf{z}_{0:T})}{q_\phi(\mathbf{z}_{0:T} \mid \mathbf{x}_{0:T})} \, d\mathbf{z}_{0:T} \\
&\geq \int q_\phi(\mathbf{z}_{0:T} \mid \mathbf{x}_{0:T}) \log \frac{p_\theta(\mathbf{x}_{0:T}, \mathbf{z}_{0:T})}{q_\phi(\mathbf{z}_{0:T} \mid \mathbf{x}_{0:T})} \, d\mathbf{z}_{0:T}
\end{aligned}$$

This defines the ELBO $\mathcal{L}(\theta, \phi; \mathbf{x}_{0:T})$.

5.1.3 Decomposition of the Objective

Substituting the factorization of the generative model into the ELBO definition:

$$\begin{aligned}
\mathcal{L}(\theta, \phi; \mathbf{x}_{0:T}) &= \mathbb{E}_{q_\phi(\mathbf{z}_{0:T}|\mathbf{x}_{0:T})}\left[\log\left(p(\mathbf{z}_{0:T}) \prod_{t=0}^{T} p_\theta(x_t|z_t)\right) - \log q_\phi(\mathbf{z}_{0:T}|\mathbf{x}_{0:T})\right] \\
&= \mathbb{E}_{q_\phi}\left[\sum_{t=0}^{T} \log p_\theta(x_t|z_t)\right] + \mathbb{E}_{q_\phi}\left[\log p(\mathbf{z}_{0:T}) - \log q_\phi(\mathbf{z}_{0:T}|\mathbf{x}_{0:T})\right]
\end{aligned}$$

Proposition 5.2 - **Reconstruction vs. Regularisation**

Let $q_\phi(z_t \mid \mathbf{x}_{0:T})$ denote the marginal of $q_\phi(\mathbf{z}_{0:T} \mid \mathbf{x}_{0:T})$ at time t.
Then the ELBO can be written as:

$$\boxed{\mathcal{L}(\theta, \phi; \mathbf{x}_{0:T}) = \underbrace{\sum_{t=0}^{T} \mathbb{E}_{q_\phi(z_t \mid \mathbf{x}_{0:T})} [\log p_\theta(x_t \mid z_t)]}_{\text{Reconstruction Term}} - \underbrace{\mathrm{KL}\Big(q_\phi(\mathbf{z}_{0:T} \mid \mathbf{x}_{0:T}) \,\|\, p(\mathbf{z}_{0:T})\Big)}_{\text{Regularisation Term}}}$$

Proof:

The second expectation is by definition:

$$\mathbb{E}_{q_\phi} [\log p(\mathbf{z}_{0:T}) - \log q_\phi(\mathbf{z}_{0:T} \mid \mathbf{x}_{0:T})] = -\mathrm{KL}\big(q_\phi(\mathbf{z}_{0:T} \mid \mathbf{x}_{0:T}) \,\|\, p(\mathbf{z}_{0:T})\big)$$

The reconstruction term decomposes into a sum over t, and each expectation can be written with respect to the marginal $q_\phi(z_t \mid \mathbf{x}_{0:T})$.

Remark (Scope):
This formulation is **fully general** and applies to **any longitudinal latent variable model**.
In the following sections, we specialise $p(\mathbf{z}_{0:T})$ and $q_\phi(\mathbf{z}_{0:T} \mid \mathbf{x}_{0:T})$ to the RLVAE setting, where trajectories are constrained by deterministic geometric flows!

5.2 Deterministic Latent Dynamics and the Path Manifold

In the RLVAE framework, longitudinal latent evolution is not modeled as a stochastic Markov process, but as a **deterministic geometric transport** over a learned Riemannian manifold.
Given an initial random seed produced by the encoder, the entire latent trajectory is **uniquely** generated through geometry-driven dynamics.
This implies that the distribution over trajectories does not admit a density with respect to the Lebesgue measure on $\mathcal{M}^{T+1}$, but is instead supported on a low-dimensional embedded submanifold.

Definition 5.3 - **Deterministic Latent Dynamics**

Let $\mathcal{M} \cong \mathbb{R}^d$ be the latent manifold.
The RLVAE assumes that the latent trajectory is generated by a **deterministic map**

$$\mathcal{T} : T^*\mathcal{M} \longrightarrow \mathcal{M}^{T+1},$$

such that for an augmented seed

$$\xi = (z_{in}, \rho_{in}),$$

the latent trajectory is given by

$$\mathbf{z}_{0:T} = \mathcal{T}(\xi)$$

This trajectory is uniquely determined by:

- A **geometric refinement operator** Φ.
- A **temporal flow** $F = \{f_t\}_{t=1}^{T}$.

5.2.1 The Transport Map

Definition 5.4 - **Geometric + Temporal Transport Operator**

The trajectory-generating map decomposes as:

$$\mathcal{T} = F \circ \Phi,$$

where:

- **Geometric refinement**

$$(z_0, \rho_0) = \Phi(z_{in}, \rho_{in}),$$

  is the RHMC-based deterministic projection onto the learned manifold.
- **Temporal propagation**

$$z_t = f_t(z_{t-1}), \quad t = 1, \dots, T,$$

  with each $f_t : \mathcal{M} \to \mathcal{M}$ a diffeomorphism.

Thus, for any seed ξ, the entire trajectory is:

$$\mathbf{z}_{0:T} = (z_0, z_1, \dots, z_T) = (z_0, f_1(z_0), f_2 \circ f_1(z_0), \dots, f_T \circ \cdots \circ f_1(z_0))$$

Proposition 5.3 - **Degeneracy of the Trajectory Measure**

The **induced distribution over trajectories** is supported on a d-dimensional immersed submanifold:

$$\Gamma = \mathcal{T}(T^*\mathcal{M}) \subset \mathcal{M}^{T+1}$$

Hence, the trajectory distribution is **singular** with respect to the **Lebesgue measure** on $\mathcal{M}^{T+1}$.

Proof:

1. **Define a canonical "path embedding" map that depends only on z_0:**
   Consider the map

$$\Psi : \mathcal{M} \to \mathcal{M}^{T+1}, \quad \Psi(z_0) = \big(z_0,\ f_1(z_0),\ f_2 \circ f_1(z_0),\ \dots,\ f_T \circ \cdots \circ f_1(z_0)\big).$$

   Each $f_t : \mathcal{M} \to \mathcal{M}$ is a diffeomorphism by assumption (smooth with smooth inverse).
   Therefore, each composite map

$$F_t = f_t \circ \cdots \circ f_1 : \mathcal{M} \to \mathcal{M}$$

   is also a diffeomorphism.
   The map Ψ is then a **smooth map** whose components are **smooth diffeomorphisms** of $\mathcal{M}$.
2. **Show that Ψ is an immersion (Jacobian of rank d):**
   Let $J_{F_t}(z_0)$ denote the Jacobian of F_t at z_0.
   The Jacobian of Ψ at z_0 is the $(T+1)d \times d$ block matrix:

$$D\Psi(z_0) = \begin{pmatrix} I_d \\ J_{F_1}(z_0) \\ J_{F_2}(z_0) \\ \vdots \\ J_{F_T}(z_0) \end{pmatrix}$$

   The top block is the identity I_d, hence the columns of $D\Psi(z_0)$ are linearly independent.
   Therefore, $\operatorname{rank}(D\Psi(z_0)) = d$ at every point: Ψ is an **immersion**.
   By standard differential geometry, the image

$$\Gamma := \Psi(\mathcal{M}) \subset \mathcal{M}^{T+1}$$

is a d-**dimensional** immersed submanifold of $\mathcal{M}^{T+1}$.

3. **Relate $\mathcal{T}$ to Ψ:**
   The full trajectory map is

$$\mathcal{T} = F \circ \Phi : T^*\mathcal{M} \to \mathcal{M}^{T+1},$$

   where $\Phi : T^*\mathcal{M} \to T^*\mathcal{M}$ is the RHMC map and

$$F : T^*\mathcal{M} \to \mathcal{M}^{T+1}, \quad F(z_0, \rho_0) = \big(z_0, f_1(z_0), \ldots, f_T \circ \cdots \circ f_1(z_0)\big)$$

   Note that F **ignores the momentum** ρ_0: it depends only on the position component z_0.
   Denote the projection

$$\pi : T^*\mathcal{M} \to \mathcal{M}, \quad \pi(z, \rho) = z$$

   Then

$$F = \Psi \circ \pi, \quad \Rightarrow \quad \mathcal{T} = F \circ \Phi = \Psi \circ \big(\pi \circ \Phi\big)$$

   Therefore,

$$\mathrm{Im}(\mathcal{T}) \subset \mathrm{Im}(\Psi) = \Gamma$$

4. **Support of the trajectory distribution:**
   The prior/posterior seeds $\xi = (z_{in}, \rho_{in})$ are sampled in $T^*\mathcal{M}$ with some density (w.r.t. the natural product measure), and then mapped via $\mathcal{T}$.
   Hence, the **support** of the induced distribution on trajectories is contained in Γ.
   Since $\mathcal{M}^{T+1} \cong \mathbb{R}^{(T+1)d}$ carries the usual Lebesgue measure and Γ has dimension $d < (T+1)d$, this **induced measure** is singular with respect to Lebesgue measure on $\mathcal{M}^{T+1}$.

5.2.2 The Trajectory Prior

Definition 5.5 - Flow-Constrained Prior

The prior over trajectories is defined as:

$$p(\mathbf{z}_{0:T}) = p_T(z_T) \prod_{t=1}^{T} \delta\big(z_t - f_t(z_{t-1})\big),$$

where:

- $p_T(z_T)$ is the precision-weighted geometric prior on the transported manifold: $p_T(z_T) = \frac{1}{Z_p}\sqrt{\det G_T^{-1}(z_T)}$,
- The Dirac distributions enforce consistency with the **deterministic flow**.

This construction ensures that only geometrically valid trajectories have **non-zero probability under the prior**.

5.2.3 The Trajectory Posterior

MAYBE CHECK THE qaug and add a proof ??

Definition 5.6 - **Flow-Constrained Posterior**

The **variational posterior** follows the same structural constraint:

$$q_\phi(\mathbf{z}_{0:T}|\mathbf{x}) = q_0(z_0|\mathbf{x}) \prod_{t=1}^{T} \delta\big(z_t - f_t(z_{t-1})\big),$$

where $q_0(z_0|\mathbf{x})$ is the marginal distribution of the initial refined latent state.
It is induced from the **augmented seed** via:

$$q_0(z_0|\mathbf{x}) = \int q_{aug}(z_0, \rho_0|\mathbf{x})\, d\rho_0,$$

with

$$q_{aug} = \Phi_\# \left[q_{in}(z_{in}|\mathbf{x})\; \pi(\rho_{in}|z_{in})\right]$$

5.2.4 Common Support and Validity of KL

Hypothesis - **Shared Support Condition**

Both prior and posterior are supported on the same trajectory manifold:

$$\mathrm{supp}(q_\phi) = \mathrm{supp}(p) = \Gamma$$

Therefore, the KL divergence is **well-defined** despite **singular densities**.

Proposition 5.4 - **Cancellation of Deterministic Constraints**

On the manifold Γ, the density ratio simplifies as:

$$\frac{q_\phi(\mathbf{z}_{0:T}|\mathbf{x})}{p(\mathbf{z}_{0:T})} = \frac{q_0(z_0|\mathbf{x})}{p_T(z_T)}$$

The Dirac terms cancel, reducing the path-wise KL to a **marginal form** — a result exploited in **Section 5.3**.

Proof:

1. **Write the prior and posterior in constrained form:**
   From Section 5.2, we defined:
   - **Trajectory prior**

   $$p(\mathbf{z}_{0:T}) = p_T(z_T) \prod_{t=1}^{T} \delta\big(z_t - f_t(z_{t-1})\big)$$

   - **Trajectory posterior**

   $$q_\phi(\mathbf{z}_{0:T}|\mathbf{x}) = q_0(z_0|\mathbf{x}) \prod_{t=1}^{T} \delta\big(z_t - f_t(z_{t-1})\big)$$

   Both are distributions on $\mathcal{M}^{T+1}$ **concentrated** on the set of valid trajectories:

   $$\Gamma = \{(z_0, \ldots, z_T) \,|\, \forall t,\; z_t = f_t(z_{t-1})\,\}$$

2. **Consider the ratio on the support Γ:**
   Let $\mathbf{z}_{0:T}$ be any trajectory satisfying the flow constraints:

$$z_t = f_t(z_{t-1}), \quad t = 1, \ldots, T$$

For such a trajectory, each Dirac term in the product evaluates to $\delta(0)$ in a distributional sense.
Formally, when computing a ratio of two distributions that are both supported on the same constraint surface, we only need to consider their **Radon–Nikodym derivatives** with respect to a common base measure on Γ.
Intuitively: the Dirac chains are identical for p and q and **cancel in the ratio** wherever both are **non-zero**.
Concretely, the "density-like" parts of each distribution are:

$$q_\phi(\mathbf{z}_{0:T}|\mathbf{x}) \propto q_0(z_0|\mathbf{x}) \prod_{t=1}^{T} \delta\big(z_t - f_t(z_{t-1})\big),$$
$$p(\mathbf{z}_{0:T}) \propto p_T(z_T) \prod_{t=1}^{T} \delta\big(z_t - f_t(z_{t-1})\big)$$

On Γ, the product of Dirac deltas is **identical** in numerator and denominator, so the ratio simplifies to:

$$\frac{q_\phi(\mathbf{z}_{0:T}|\mathbf{x})}{p(\mathbf{z}_{0:T})} = \frac{q_0(z_0|\mathbf{x})}{p_T(z_T)}, \quad \text{for } \mathbf{z}_{0:T} \in \Gamma$$

3. **Outside the support:**
   For any $\mathbf{z}_{0:T} \notin \Gamma$, at least one constraint $\delta(z_t - f_t(z_{t-1}))$ evaluates to zero in both prior and posterior. Then:

$$q_\phi(\mathbf{z}_{0:T}|\mathbf{x}) = p(\mathbf{z}_{0:T}) = 0,$$

   and these points **do not** contribute to any KL integral (they lie outside the common support).
4. **Conclusion:**
   Hence, almost everywhere with respect to the common support measure on Γ, we have:

$$\frac{q_\phi(\mathbf{z}_{0:T}|\mathbf{x})}{p(\mathbf{z}_{0:T})} = \frac{q_0(z_0|\mathbf{x})}{p_T(z_T)}$$

5.3 Reduction of the Path-Wise KL to Final-State KL

The variational objective requires computing the **Kullback-Leibler divergence** between distributions defined over the high-dimensional trajectory space $\mathcal{Z}^{T+1} = \mathcal{M}^{T+1}$.
At first sight, this appears to require integration over a $(T+1)d$-dimensional space.
However, as shown in **Section 5.2**, both the prior and posterior are concentrated on the same deterministic path manifold induced by the flow dynamics.
We now prove that this structural degeneracy causes the KL divergence **over trajectories** to collapse exactly to a KL divergence between the **final-time marginals**.

Proposition 5.5 - Collapse of a Path-Wise KL under Deterministic Flow
Let $p(\mathbf{z}_{0:T})$ and $q_\phi(\mathbf{z}_{0:T}|\mathbf{x})$ be two distributions supported on the same sub-manifold of valid trajectories defined by the invertible flow maps $\{f_t\}_{t=1}^{T}$.
The KL divergence over the full trajectory is equivalent to the KL divergence at the final state:

$$\mathrm{KL}\Big(q_\phi(\mathbf{z}_{0:T}|\mathbf{x}) \parallel p(\mathbf{z}_{0:T})\Big) = \mathrm{KL}\Big(q_T(z_T|\mathbf{x}) \parallel p_T(z_T)\Big)$$

Proof:

From **Section 5.2**, both distributions factorize as singular measures supported on Γ:

$$p(\mathbf{z}_{0:T}) = p_T(z_T) \prod_{t=1}^{T} \delta\big(z_t - f_t(z_{t-1})\big),$$
$$q_\phi(\mathbf{z}_{0:T}|\mathbf{x}) = q_0(z_0|\mathbf{x}) \prod_{t=1}^{T} \delta\big(z_t - f_t(z_{t-1})\big)$$

By **Proposition 5.4**, for any trajectory in Γ, the **Radon–Nikodym** derivative simplifies to

$$\frac{q_\phi(\mathbf{z}_{0:T}|\mathbf{x})}{p(\mathbf{z}_{0:T})} = \frac{q_0(z_0|\mathbf{x})}{p_T(z_T)}$$

Therefore, the KL divergence becomes:

$$\begin{aligned}\mathrm{KL}(q_\phi \| p) &= \int_\Gamma q_\phi(\mathbf{z}_{0:T}|\mathbf{x}) \log \frac{q_\phi(\mathbf{z}_{0:T}|\mathbf{x})}{p(\mathbf{z}_{0:T})}\, d\mu_\Gamma \\ &= \int_\Gamma q_\phi(\mathbf{z}_{0:T}|\mathbf{x}) \log \frac{q_0(z_0|\mathbf{x})}{p_T(z_T)}\, d\mu_\Gamma,\end{aligned}$$

where $d\mu_\Gamma$ is the induced measure on the manifold Γ.
Since z_T is a deterministic function of z_0, integrating over Γ corresponds to integrating over z_T with respect to its marginal distribution q_T, yielding:

$$\mathrm{KL}(q_\phi \| p) = \int_\mathcal{M} q_T(z_T|\mathbf{x}) \log \frac{q_T(z_T|\mathbf{x})}{p_T(z_T)}\, dz_T = \mathrm{KL}(q_T \| p_T)$$

This proves the result.

Corollary 5.1 - Endpoint Regularisation Principle
The **regularization term** of the ELBO depends solely on the terminal latent state:

$$\mathcal{L}_{reg} = \mathbb{E}_{z_T \sim q_T}[\log q_T(z_T|\mathbf{x}) - \log p_T(z_T)]$$

Thus, geometric consistency is enforced exclusively at the **final transported configuration**.

This result is central to the RLVAE framework:

- The complexity of **enforcing geometric consistency across time** is replaced by a **single constraint at the final state**.
- The entire trajectory is shaped implicitly through **deterministic flow**, but penalized **only at its endpoint**.
- This structure enables a principled ***balance*** between **temporal flexibility** and **geometric regularity**.

5.4 Explicit Derivation of the Expansion Drive

Having reduced the path-wise divergence to the final state marginals (Proposition 5.5), we now explicitly analyse the geometric structure of the term $\log q_T(z_T|\mathbf{x}) - \log p_T(z_T)$ by pulling it back to the initial reference frame at $t = 0$.
This reveals a fundamental structural asymmetry: **probability densities contract under expansion, while precision-weighted geometric densities expand**, generating an intrinsic drive towards volume inflation.

Hypothesis - Regularity Conditions

We assume:

- The flow $F : \mathcal{M} \to \mathcal{M}$ is a **diffeomorphism**.
- The Jacobian $J_F(z)$ is **non-singular everywhere**.
- The metric field $G(z)$ is **smooth** and **positive-definite**.
- The base posterior q_0 is **absolutely continuous** w.r.t. Lebesgue measure.

Lemma 5.1- Posterior Entropy Contraction
Under a diffeomorphism $F : \mathcal{M} \to \mathcal{M}$, the **log-density** of the **pushforward distribution** $q_T = F_\# q_0$ is given by:

$$\log q_T(F(z_0)|\mathbf{x}) = \log q_0(z_0|\mathbf{x}) - \log|\det J_F(z_0)|$$

where $J_F(z_0) = \frac{\partial F}{\partial z_0}$ is the Jacobian of the transformation.

Proof:

By the change of variables formula for probability density functions, the conservation of probability mass requires:

$$q_T(z_T)\,|dz_T| = q_0(z_0)\,|dz_0|$$

Substituting the volume change $|dz_T| = |\det J_F(z_0)|\,|dz_0|$:

$$q_T(F(z_0)) = \frac{q_0(z_0)}{|\det J_F(z_0)|}$$

Taking the logarithm yields the lemma.
This term represents the **standard entropy change**: if the flow **expands** the volume ($|\det J_F| > 1$), the probability density at a point must **decrease** to maintain **normalization**.

Lemma 5.2 - Geometric Precision Expansion
Let $\mathcal{M}$ be a manifold equipped with a Riemannian metric that is transported by the flow F (i.e., $G_T = (F^{-1})^{G_0}$).
We adopt a **Precision-First geometry** where $p_T(z) \propto \sqrt{\det G_T^{-1}(z)}$.
Since the precision matrix transforms contravariantly ($G_T^{-1} = JG_0^{-1}J^\top$), its determinant scales by $(\det J)^2$. Thus:

$$\log p_T(z_T) = \frac{1}{2}\log\det G_0^{-1}(z_0) + \log|\det J_F(z_0)| + C$$

Expansion of the space **increases** the **volume of precision**, rewarding the prior.

Proof:

Let $P(z) = G^{-1}(z)$ denote the precision matrix.
The Riemannian metric G is a covariant $(0, 2)$-tensor.
Under the diffeomorphism $z_T = F(z_0)$, the metric transforms via the pullback of the inverse map:

$$G_T(z_T) = (J_F^{-1})^\top G_0(z_0)(J_F^{-1}),$$

where $J_F = \frac{\partial z_T}{\partial z_0}$ is the Jacobian matrix of the flow.
The precision matrix $P_T = G_T^{-1}$ is the inverse of the metric.
Taking the matrix inverse of the transformation law above implies that P transforms as a contravariant $(2, 0)$-tensor:

$$P_T(z_T) = \left((J_F^{-1})^\top G_0(J_F^{-1})\right)^{-1} = J_F\,G_0^{-1}\,J_F^\top = J_F\,P_0(z_0)\,J_F^\top$$

We now compute the determinant of the transported precision matrix:

$$\begin{aligned}
\det P_T(z_T) &= \det\left(J_F(z_0)\,P_0(z_0)\,J_F(z_0)^\top\right) \\
&= \det(J_F)\cdot\det(P_0(z_0))\cdot\det(J_F^\top) \\
&= (\det J_F(z_0))^2 \det P_0(z_0)
\end{aligned}$$

The prior density is defined proportional to the square root of the precision determinant:

$$p_T(z_T) = \frac{1}{\mathcal{Z}_p}\sqrt{\det P_T(z_T)}$$

Taking the logarithm:

$$\begin{aligned}
\log p_T(z_T) &= \log\left(\frac{1}{\mathcal{Z}_p}\sqrt{(\det J_F)^2\det P_0(z_0)}\right) \\
&= \frac{1}{2}\log\left((\det J_F)^2\right) + \frac{1}{2}\log\det G_0^{-1}(z_0) - \log\mathcal{Z}_p \\
&= \log|\det J_F(z_0)| + \frac{1}{2}\log\det G_0^{-1}(z_0) + C
\end{aligned}$$

Thus, flow expansion ($\log|\det J| > 0$) strictly increases the log-probability of the geometric prior.

Unlike the posterior density (Lemma 5.1), which decreases upon expansion, the geometric prior density **increases** upon **expansion**. Stretching the latent space increases the **volume of precision**, effectively creating a probability reward for expansion.

Proposition 5.6 - The Expansion Drive Formula
The **Kullback-Leibler divergence** between the transported posterior and the transported precision-weighted prior is given by:

$$\mathrm{KL}(q_T||p_T) = \mathbb{E}_{z_0\sim q_0}\left[\log\frac{q_0(z_0|\mathbf{x})}{p_0(z_0)} - 2\log|\det J_F(z_0)|\right]$$

Consequently, **minimizing** the KL divergence strictly implies **maximizing** the volume **expansion** of the flow.

Proof:

We substitute the results from Lemma 5.1 and Lemma 5.2 into the definition of the KL divergence:

$$\begin{aligned}\log\frac{q_T(z_T|\mathbf{x})}{p_T(z_T)} &= \log q_T(z_T|\mathbf{x}) - \log p_T(z_T)\\ &= \Big(\log q_0(z_0|\mathbf{x}) - \log|\det J_F|\Big) - \Big(\log p_0(z_0) + \log|\det J_F|\Big)\\ &= \log q_0(z_0|\mathbf{x}) - \log p_0(z_0) - 2\log|\det J_F(z_0)|\end{aligned}$$

Taking the expectation over $z_T \sim q_T$ (which is equivalent to $z_0 \sim q_0$ by the ***Law of the Unconscious Statistician***) yields the proposition.

Remark (Stability):
The term $-2\log|\det J_F|$ acts as an intrinsic **Expansion Drive**.
In the absence of regularization, the optimization objective is unbounded from below with respect to the flow parameters: the model can arbitrarily lower the loss by scaling the Jacobian determinant towards infinity ($\det J \to \infty$).
This mathematical property **necessitates the introduction of an explicit regularization term** $\mathcal{L}_{reg} = \lambda_{flow}\sum\log|\det J_f|$ in the final objective (Section 5.6), with $\lambda_{flow} \geq 2$ required to counteract the expansion drive and ensure a compact manifold representation.

5.5 Numerical Corrections for Riemannian Hamiltonian Monte Carlo

In the ideal, continuous-time construction of Stage C, the RHMC refinement operator Φ is a Hamiltonian flow on the cotangent bundle $T^*\mathcal{M}$, preserving phase-space volume and Hamiltonian energy.
In that setting, the refined state z_0 would follow an exact geometric posterior determined solely by the base proposal and the Hamiltonian.
In practice, Φ is implemented via a **discrete generalized leapfrog integrator** on a Riemannian manifold.
Explicit integrators are only approximately symplectic and only approximately energy-conserving. As a result:

- The joint density of (z_0, ρ_0) differs from the ideal Hamiltonian pushforward.
- We must **account explicitly** for the Jacobian of Φ and the change in kinetic energy to maintain a valid variational bound.

We now formalise this correction using the change-of-variables formula on the augmented phase space.

5.5.1 The Augmented Phase Space

We treat the geometric refinement as a deterministic mapping on the augmented phase space (position $\times$ momentum).

Definition 5.6 - **Augmented Proposal**

Let the initial seed $(z_{in}, \rho_{in}) \in T^*\mathcal{M}$ be drawn from the product distribution:

$$(z_{in}, \rho_{in}) \sim q_{in}(z|\mathbf{x})\,\pi(\rho|z),$$

where the **base encoder proposal** is

$$q_{in}(z|\mathbf{x}) = \mathcal{N}\big(z;\, \mu(\mathbf{x}), \alpha\, G_0^{-1}(\mu(\mathbf{x}))\big),$$

and the **conditional momentum distribution** is

$$\pi(\rho|z) = \mathcal{N}\big(\rho;\, 0,\, G_0(z)\big)$$

The RHMC refinement operator $\Phi : T^*\mathcal{M} \to T^*\mathcal{M}$ is defined as the composition of L generalized leapfrog steps:

$$(z_0, \rho_0) = \Phi(z_{in}, \rho_{in})$$

→ Conditioned on (z_{in}, ρ_{in}), the refined state (z_0, ρ_0) is **deterministic**.

5.5.2 The Joint Density Pushforward

Definition 5.7 - **Augmented Proposal Measure**

On the augmented space $T^*\mathcal{M} \cong \mathbb{R}^{2d}$, we define the **augmented proposal density**

$$q_{\text{aug}}(z_{in}, \rho_{in} \mid \mathbf{x}) \;=\; q_{in}(z_{in} \mid \mathbf{x})\,\pi(\rho_{in} \mid z_{in}),$$

where

$$q_{in}(z_{in} \mid \mathbf{x}) = \mathcal{N}\big(z_{in}; \mu(\mathbf{x}), \alpha G_0^{-1}(\mu(\mathbf{x}))\big) \quad \text{and} \quad \pi(\rho_{in} \mid z_{in}) = \mathcal{N}\big(\rho_{in}; 0, G_0(z_{in})\big)$$

Let $\Phi : T^*\mathcal{M} \to T^*\mathcal{M}$ be the deterministic RHMC map (composition of L leapfrog steps), $(z_0, \rho_0) = \Phi(z_{in}, \rho_{in})$.
We define the **refined joint measure** as the pushforward of the augmented proposal:

$$Q_{\text{joint}} \;\equiv\; \Phi_{\#} Q_{\text{aug}},$$

i.e. for any measurable set $A \subset T^*\mathcal{M}$,

$$Q_{\text{joint}}(A) \;=\; Q_{\text{aug}}\big(\Phi^{-1}(A)\big)$$

If this measure admits a **density** w.r.t. Lebesgue, we denote it by

$$q_{\text{joint}}(z_0, \rho_0 \mid \mathbf{x})$$

So:

- q_{aug} is the **density at the input of RHMC** (z_{in}, ρ_{in}).
- q_{joint} is the **density at the output** (z_0, ρ_0), induced by Φ.
- The relation between them is exactly the usual pushforward relation.

We now compute the joint density of the refined state using a standard change-of-variables argument.

Hypothesis 5.2 - Regularity of the Integrator

We assume that the numerical RHMC map Φ is a diffeomorphism on its domain, with Jacobian:

$$J_\Phi(z_{in}, \rho_{in}) = \frac{\partial(z_0, \rho_0)}{\partial(z_{in}, \rho_{in})}$$

everywhere invertible and smoothly varying.

Proposition 5.6 - Joint Density of the Refined State

Under Hypothesis 5.2, the joint density of the refined state (z_0, ρ_0) is:

$$\log q_{\text{joint}}(z_0, \rho_0|\mathbf{x}) = \log q_{in}(z_{in}|\mathbf{x}) + \log \pi(\rho_{in}|z_{in}) - \log \left| \det J_\Phi(z_{in}, \rho_{in}) \right|$$

Proof:

We work in the augmented space $T^*\mathcal{M} \cong \mathbb{R}^{2d}$ with coordinates:

$$u = (z_{in}, \rho_{in}), \quad v = (z_0, \rho_0) = \Phi(u)$$

Pushforward at the level of measures:

By definition of the pushforward measure $Q_{\text{joint}} = \Phi_\# Q_{\text{aug}}$, for any measurable set $A \subset T^*\mathcal{M}$,

$$Q_{\text{joint}}(A) = Q_{\text{aug}}\left(\Phi^{-1}(A)\right) = \int_{\Phi^{-1}(A)} q_{\text{aug}}(u \mid \mathbf{x})\, du$$

If Q_{joint} admits a density q_{joint} w.r.t. Lebesgue, we must have

$$Q_{\text{joint}}(A) = \int_A q_{\text{joint}}(v \mid \mathbf{x})\, dv$$

Use a test function (standard trick):

To identify the density, take any bounded measurable test function $\varphi : T^*\mathcal{M} \to \mathbb{R}$.

By definition of **expectation** under the pushforward,

$$\int \varphi(v)\, q_{\text{joint}}(v \mid \mathbf{x})\, dv = \int \varphi\big(\Phi(u)\big)\, q_{\text{aug}}(u \mid \mathbf{x})\, du$$

We want to rewrite the right-hand side as an integral over v.

Change of variables using the diffeomorphism:

Since Φ is a diffeomorphism with inverse Φ^{-1}, we can use the change-of-variables formula with

$$v = \Phi(u) \quad \Longleftrightarrow \quad u = \Phi^{-1}(v)$$

The Jacobian determinant of the inverse map is

$$J_{\Phi^{-1}}(v) = \frac{\partial u}{\partial v}, \quad |\det J_{\Phi^{-1}}(v)| = \frac{1}{|\det J_\Phi(u)|},$$

where $u = \Phi^{-1}(v)$.

The change-of-variables formula yields:

$$\int \varphi\big(\Phi(u)\big)\, q_{\text{aug}}(u \mid \mathbf{x})\, du = \int \varphi(v)\, q_{\text{aug}}\big(\Phi^{-1}(v) \mid \mathbf{x}\big)\ |\det J_{\Phi^{-1}}(v)|\ dv$$

Identify the density by uniqueness:

We thus have, for all test functions φ,

$$\int \varphi(v)\, q_{\text{joint}}(v \mid \mathbf{x})\, dv = \int \varphi(v)\, q_{\text{aug}}\big(\Phi^{-1}(v) \mid \mathbf{x}\big)\ |\det J_{\Phi^{-1}}(v)|\ dv$$

By uniqueness of densities (two densities that give the same integral for all test functions must coincide almost everywhere), we must have

$$q_{\text{joint}}(v \mid \mathbf{x}) = q_{\text{aug}}\big(\Phi^{-1}(v) \mid \mathbf{x}\big)\ |\det J_{\Phi^{-1}}(v)| \quad \text{a.e. in } v$$

Rewriting in terms of original variables:

$$v = (z_0, \rho_0), \quad u = (z_{in}, \rho_{in}) = \Phi^{-1}(z_0, \rho_0),$$

and recalling that $q_{\text{aug}}(u \mid \mathbf{x}) = q_{in}(z_{in} \mid \mathbf{x})\, \pi(\rho_{in} \mid z_{in})$, we obtain:

$$q_{\text{joint}}(z_0, \rho_0 \mid \mathbf{x}) = q_{in}(z_{in} \mid \mathbf{x})\, \pi(\rho_{in} \mid z_{in}) \left|\det J_{\Phi^{-1}}(z_0, \rho_0)\right|$$

Log form and relation to J_Φ:

Taking the logarithm:

$$\log q_{\text{joint}}(z_0, \rho_0 \mid \mathbf{x}) = \log q_{in}(z_{in} \mid \mathbf{x}) + \log \pi(\rho_{in} \mid z_{in}) + \log\left|\det J_{\Phi^{-1}}(z_0, \rho_0)\right|$$

Since $|\det J_{\Phi^{-1}}| = 1/|\det J_\Phi|$, this is equivalently:

$$\log q_{\text{joint}}(z_0, \rho_0 \mid \mathbf{x}) = \log q_{in}(z_{in} \mid \mathbf{x}) + \log \pi(\rho_{in} \mid z_{in}) - \log\left|\det J_\Phi(z_{in}, \rho_{in})\right|$$

This is the desired result.

5.5.3 Volume Distortion (Δ_{vol})

In ideal (continuous) RHMC, the Hamiltonian flow is symplectic and exactly volume-preserving in phase space, so $|\det J_\Phi| = 1$.
For practical, explicit integrators on a manifold, this is only approximately true.

Definition 5.8 - Volume Distortion

We define the Volume Distortion term as:

$$\Delta_{vol} \equiv \log|\det J_\Phi(z_{in}, \rho_{in})|$$

Remark (Symplectic Limit):
If Φ were exactly symplectic and volume-preserving, we would have $\Delta_{vol} \equiv 0$.
In our implementation, Δ_{vol} is small but non-zero and must be included for mathematical correctness.
With this notation, Proposition 5.6 rewrites as:

$$\log q_{\text{joint}}(z_0, \rho_0|\mathbf{x}) = \log q_{in}(z_{in}|\mathbf{x}) + \log \pi(\rho_{in}|z_{in}) - \Delta_{vol}$$

5.5.4 Kinetic Drift (Δ_{kin}) via an Auxiliary-Variable Bound

Our goal is to obtain an approximate expression for the marginal density $q_0(z_0 \mid \mathbf{x})$, starting from the joint refined density $q_{\text{joint}}(z_0, \rho_0 \mid \mathbf{x})$.
By definition,

$$q_0(z_0 \mid \mathbf{x}) = \int q_{\text{joint}}(z_0, \rho_0 \mid \mathbf{x})\, d\rho_0,$$

but this integral is generally intractable.
We therefore use an **auxiliary-variable variational trick**: we treat ρ_0 as a latent variable with a chosen “auxiliary posterior” and derive a lower bound on $\log q_0(z_0 \mid \mathbf{x})$.

Proposition 5.7 - Auxiliary-Variable Lower Bound

Let $r(\rho_0 \mid z_0)$ be any probability density on ρ_0 conditioned on z_0.
Then:

$$\log q_0(z_0 \mid \mathbf{x}) \;\geq\; \mathbb{E}_{\rho_0 \sim r(\cdot|z_0)}\Big[\log q_{\text{joint}}(z_0, \rho_0 \mid \mathbf{x}) - \log r(\rho_0 \mid z_0)\Big]$$

Proof:

Start from the definition:

$$q_0(z_0 \mid \mathbf{x}) = \int q_{\text{joint}}(z_0, \rho_0 \mid \mathbf{x})\, d\rho_0$$

Multiply and divide by an arbitrary conditional density $r(\rho_0 \mid z_0)$:

$$q_0(z_0 \mid \mathbf{x}) = \int r(\rho_0 \mid z_0) \frac{q_{\text{joint}}(z_0, \rho_0 \mid \mathbf{x})}{r(\rho_0 \mid z_0)}\, d\rho_0$$

Taking the logarithm and applying **Jensen's inequality** (log of an expectation):

$$\begin{aligned} \log q_0(z_0 \mid \mathbf{x}) &= \log \mathbb{E}_{\rho_0 \sim r(\cdot|z_0)}\left[\frac{q_{\text{joint}}(z_0, \rho_0 \mid \mathbf{x})}{r(\rho_0 \mid z_0)}\right] \\ &\geq \mathbb{E}_{\rho_0 \sim r(\cdot|z_0)}\left[\log q_{\text{joint}}(z_0, \rho_0 \mid \mathbf{x}) - \log r(\rho_0 \mid z_0)\right] \end{aligned}$$

This is the desired inequality.

In our case, a natural choice of auxiliary density is the **Riemannian Gaussian** used for the momentum proposal:

Definition 5.9 - Auxiliary Momentum Posterior

We choose:

$$r(\rho_0 \mid z_0) \;\equiv\; \pi(\rho_0 \mid z_0) = \mathcal{N}\big(\rho_0; 0, G_0(z_0)\big),$$

i.e. the same local Riemannian Gaussian that defines the initial momentum distribution, but evaluated at the final state (z_0, ρ_0).

With this choice, the lower bound becomes

$$\log q_0(z_0 \mid \mathbf{x}) \gtrsim \underbrace{\log q_{\text{joint}}(z_0, \rho_0 \mid \mathbf{x})}_{\text{from Prop. 5.6}} \;-\; \log \pi(\rho_0 \mid z_0),$$

where the "$\gtrsim$" reminds us this is a **lower bound** (we will treat it as a working approximation and denote it by $\log qeff$ below).

Using the expression from Proposition 5.6:

$$\log q_{\text{joint}}(z_0, \rho_0 \mid \mathbf{x}) = \log q_{in}(z_{in} \mid \mathbf{x}) + \log \pi(\rho_{in} \mid z_{in}) - \Delta_{vol}$$

with

$$\Delta_{vol} \equiv \log\big|\det J_\Phi(z_{in}, \rho_{in})\big|,$$

we obtain

$$\begin{aligned} \log q_0(z_0 \mid \mathbf{x}) &\gtrsim \Big[\log q_{in}(z_{in} \mid \mathbf{x}) + \log \pi(\rho_{in} \mid z_{in}) - \Delta_{vol}\Big] - \log \pi(\rho_0 \mid z_0) \\ &= \log q_{in}(z_{in} \mid \mathbf{x}) + \big[\log \pi(\rho_{in} \mid z_{in}) - \log \pi(\rho_0 \mid z_0)\big] - \Delta_{vol} \end{aligned}$$

This motivates the following definition.

Definition 5.10 - Kinetic Drift

We define the **Kinetic Drift** term as:

$$\Delta_{kin} \equiv \log \pi(\rho_{in}|z_{in}) - \log \pi(\rho_0|z_0)$$

It measures the **change** in the **momentum log-density** (and hence in kinetic energy) between the **initial** and **final** states of the integrator.

Interpretation:

- For **an *ideal,* energy-preserving** RHMC integrator with exact Hamiltonian flow, the joint Hamiltonian $H(z, \rho)$ would be **conserved** and the momentum distribution at the end would match the initial one up to the change in potential; in that case Δ_{kin} would capture only the ***potential/energy reshuffling***.
- For **a numerical integrator**, Δ_{kin} also absorbs **discretization errors** in kinetic energy and **deviations** from the exact target energy level.

5.5.5 Effective Log-Density

We can now summarize the approximate log-density of the refined position z_0 that enters the longitudinal flow.
Starting from the auxiliary-variable lower bound and our definitions:

$$\log q_0(z_0 \mid \mathbf{x}) \ \gtrsim \ \log q_{in}(z_{in} \mid \mathbf{x}) + \Delta_{kin} - \Delta_{vol},$$

with

$$\Delta_{kin} = \log \pi(\rho_{in} \mid z_{in}) - \log \pi(\rho_0 \mid z_0), \qquad \Delta_{vol} = \log \left| \det J_\Phi(z_{in}, \rho_{in}) \right|$$

This motivates the *effective* posterior log-density used in the ELBO:

Definition 5.11 - Effective Refined Density

We define the **effective log-density** of the refined state z_0 as

$$\boxed{\log q_{\text{eff}}(z_0 \mid \mathbf{x}) \ \equiv \ \log q_{in}(z_{in} \mid \mathbf{x}) + \Delta_{kin} - \Delta_{vol}}$$

where $(z_0, \rho_0) = \Phi(z_{in}, \rho_{in})$.

This expression is what actually appears inside the KL term once we:

1. Start from the **encoder proposal** $q_{in}(z_{in} \mid \mathbf{x})$.
2. Lift into **phase space** with $\pi(\rho_{in} \mid z_{in})$.
3. Apply the **RHMC map** Φ.
4. Correct for:

- **Volume Distortion** Δ_{vol} (non–volume-preserving leapfrog on the manifold).
- **Kinetic Drift** Δ_{kin} (change in momentum density).

Practical consequence:

- Gradients of the loss w.r.t. **encoder parameters** ϕ flow through $\log q_{in}$.
- Gradients w.r.t. **metric / RHMC parameters** flow through Δ_{kin} (via G_0, the kinetic energy, and momentum distributions).
- Gradients w.r.t. the **integrator path** flow through Δ_{vol} (via the Jacobian of Φ).

This makes the RHMC refinement an explicit, differentiable "geometric preconditioning layer" in front of the longitudinal flow, with a fully controlled contribution to the variational objective.

5.6 The Final Operational Objective and Manifold Stability

We now assemble the components derived in the preceding sections to formulate the final objective function optimized during training. We show that the **intrinsic geometry** of the precision-weighted prior induces an ***expansion instability in the latent flows***, and that an explicit stabilizing term is therefore required to ensure compactness and numerical well-posedness.

5.6.1 The Pure Geometric Regularizer

Recall that the ELBO regularization term reduces to a KL divergence at the final state:

$$\mathcal{L}_{reg} = \mathrm{KL}(q_T \,||\, p_T)$$

Using:

- the **Expansion Drive** (**Proposition 4.5**),
- the **RHMC corrections** (**Section 5.5**),
- and the **effective refined density** $q_{\mathrm{eff}}(z_0)$,

we define the intrinsic geometric KL term:

Definition 5.12 - The Unregularized Geometric Loss
The **geometric KL term** induced by the precision-weighted prior is:

$$\boxed{\mathcal{KL}_{geo}(\phi) = \mathbb{E}_{\substack{z_{in} \sim q_{in} \\ \rho_{in} \sim \pi}} \left[\underbrace{\log q_{in}(z_{in}|\mathbf{x}) + \Delta_{kin} - \Delta_{vol}}_{\text{Effective posterior density}} - \underbrace{\frac{1}{2} \log \det G_0^{-1}(z_0)}_{\text{Base precision}} - \underbrace{2 \sum_{t=1}^{T} \log |\det J_{f_t}(z_{t-1})|}_{\text{Expansion drive}} \right]}$$

This form arises directly from the geometry and contains **no heuristic modification**.

5.6.2 Instability of the Pure Objective

This intrinsic objective is **mathematically inconsistent** with compact latent representations.

Proposition 5.8 - Divergence of the Unconstrained Flow
Consider the family of flow transformations $\mathcal{F}$.
If the flow parameters are optimized **solely to minimize** $\mathcal{KL}_{geo}(\phi)$, the objective is **unbounded from below**.

Proof:

Consider a simple scaling flow $f_t(z) = cz$ with scalar expansion factor $c > 1$.
Then:

$$\det J_{f_t} = c^d \quad \Rightarrow \quad -2 \log |\det J_{f_t}| = -2d \log c$$

The remaining terms in $\mathcal{KL}_{geo}$ are independent of c. Therefore,

$$\mathcal{KL}_{geo} \to -\infty \quad \text{as} \quad c \to \infty$$

Hence, the objective is minimized by arbitrarily **expanding the latent volume**, leading to **manifold explosion**.

This is not a numerical pathology, but a **structural property of the precision-weighted prior itself**.

5.6.3 Stabilized Objective via Flow Regularization

To counteract this intrinsic expansion drive, we introduce an **explicit penalization of volume growth**.

Definition 5.13 — Stabilized Regularization Term

We define the stabilized regularization term as:

$$\boxed{\mathcal{L}_{reg}(\mathbf{x}) = \mathbb{E}_{\substack{z_{in} \sim q_{in} \\ \rho_{in} \sim \pi}} \left[\log q_{in}(z_{in}|\mathbf{x}) + \Delta_{kin} - \Delta_{vol} - \frac{1}{2} \log \det G_0^{-1}(z_0) + \lambda_{flow} \sum_{t=1}^{T} \log |\det J_{f_t}(z_{t-1})| \right]}$$

Here, λ_{flow} is a **tunable Lagrange multiplier enforcing volume control**.

The final training objective becomes:

Definition 5.14 — Final Operational Loss

$$\boxed{\mathcal{L}_{total}(\mathbf{x}) = \sum_{t=0}^{T} \mathbb{E}_{z_t}\left[\log p_\theta(x_t|z_t)\right] \; - \; \beta\, \mathcal{L}_{reg}(\mathbf{x})}$$

5.6.4 Condition for Manifold Stability

The competition between intrinsic expansion and enforced compression yields:

Corollary 5.2 - Compactness Criterion

Let the net flow coefficient be $\kappa = \lambda_{flow} - 2$.

- If $\kappa > 0$: the manifold is penalized for expansion → stable, compact geometry.
- If $\kappa = 0$: neutral equilibrium → marginal stability.
- If $\kappa < 0$: expansion is rewarded → divergence.

Thus, **compactness requires**:

$$\boxed{\lambda_{flow} > 2}$$

5.6.5 Interpretation

The stabilized regularizer can be seen as a geometric Lagrangian:

$$\mathcal{L}_{reg} = \text{Entropy} + \text{Kinetic correction} - \text{Precision attraction} + \text{Controlled volume deformation}$$

The model therefore learns flows that:

- ***Deform latent space*** only when **necessary** for reconstruction.
- Remain anchored to **high-precision geometric regions**,
- Avoid **unbounded** inflation,
- Produce **compact, biologically meaningful trajectories**.

Implementation Note

In practice, we use:

$$\lambda_{flow} \gg 2 \quad (\text{typically } \lambda_{flow} \in [10, 30])$$

to enforce strict geometric stability.

This deliberately **breaks strict flow-invariance**, but guarantees:

- **bounded** Jacobians,
- **stable** training,
- **coherent manifold geometry**.

5.7 Interpretation: The Battle of Forces in Latent Space

The decomposition of the operational objective $\mathcal{L}_{total}$ reveals that the training of the RLVAE is not merely a **reconstruction task**, but a **dynamic equilibrium problem**.

The latent trajectory emerges from the ***tension*** between **four distinct geometric forces** acting on the probability density.

We analyse these forces based on their contribution to the negative ELBO (the loss function to be minimized).

The following section provides an interpretative reading of the variational objective in terms of competing geometric pressures acting on the latent space. These ***forces*** should be understood as ***gradients of the corresponding energy contributions***, rather than physical forces in a strict mechanical sense.

This viewpoint offers an intuitive lens for understanding stability and deformation mechanisms in the RLVAE.

5.7.1 The Entropy Force (Dispersion)

$$\mathcal{F}_{entropy} = \mathbb{E}[\log q_{in}(z_{in}|\mathbf{x})]$$

Nature: **Repulsive** / **Expansive**.
Mechanism: This term penalizes **over-confidence**.
It creates an **osmotic pressure** that encourages the base posterior to spread out and ***occupy volume*** in the latent space.
Interaction: This force opposes the **Precision Attraction** and the **Reconstruction terms**, which generally favor point-estimates. It ensures the model retains a probabilistic character, capturing the **epistemic uncertainty** of the encoder.

5.7.2 The Precision Attraction (Gravitation)

$$\mathcal{F}_{prec} = \mathbb{E}\left[-\frac{1}{2}\log\det G_0^{-1}(z_0)\right]$$

Nature: **Attractive** / **Compressive**.
Mechanism: Since the prior probability mass is concentrated in regions of high precision ($p \propto \sqrt{\det G^{-1}}$), minimizing the loss is equivalent to **maximizing this precision**.
This force acts as a **gravitational pull**, dragging the initial latent state z_0 towards the **centroids** of the learned manifold.
Interaction: This force anchors the trajectory. It **prevents** the latent code from **drifting** into the low-precision/empty spaces where the manifold structure is undefined.

5.7.3 The Expansion Drive (Inflation)

$$\mathcal{F}_{expand} = \mathbb{E}\left[-2\sum_{t=1}^{T}\log|\det J_{f_t}|\right]$$

Nature: **Expansive** / **Explosive**.
Mechanism: The geometry of the **precision-weighted prior** creates a paradox: **stretching** the latent space increases the **volume of precision**, thereby increasing the prior probability. This creates an intrinsic, **unbounded drive** for the flows to expand the latent volume ($|\det J| \to \infty$).
Interaction: If left unchecked, this force causes **Manifold Explosion**, where the latent variance grows exponentially to minimize the KL divergence, destroying the local structure required for reconstruction.

5.7.4 The Manifold Constraint (Compression)

$$\mathcal{F}_{constrain} = \mathbb{E}\left[\lambda_{flow}\sum_{t=1}^{T}\log|\det J_{f_t}|\right]$$

Nature: **Compressive** / **Elastic**.
Mechanism: The explicit regularization weight λ_{flow} introduces a **penalty** proportional to the volume expansion.
Interaction: This force **counteracts** the **Expansion Drive.** The net force acting on the flow volume is proportional to $(\lambda_{flow} - 2)$.
When $\lambda_{flow} > 2$, the net force is compressive, the flows are penalized for expanding space unless that expansion provides a significant gain in reconstruction accuracy (likelihood).

5.7.5 Summary of Equilibrium

The final trajectory $\mathbf{z}_{0:T}$ is the result of the system settling into a minimum energy state defined by these opposing pressures:

- **Initialization (z_0):** The encoder places the seed where the **Entropy Force** (uncertainty) balances the **Precision Attraction** (manifold structure).
  The sample settles near a centroid but retains enough variance to be generative.
- **Propagation ($z_{1:T}$):** The flows deform this seed to match the temporal evolution of the data.
  The extent of this deformation is governed by the **Net Flow Force**: the flows stretch space to lower reconstruction error, but are tightly bound by the **Manifold Constraint** to prevent runaway expansion.
  This formulation guarantees that the learned trajectories are not only **accurate** (low reconstruction error) but also **geometrically compact** and structurally supported by the learned metric field.

6 - Reconstruction Term

While the previous parts established the geometric and dynamical structure of latent trajectories, the model must still account for the **observable data itself**: medical images evolving through time.
This is achieved through a probabilistic **reconstruction mechanism** linking each latent state z_t to its corresponding observation x_t.
The role of the likelihood term is **twofold**:

1. **Data fidelity:** Ensure reconstructed images remain **faithful** to **medical reality**.
2. **Constraint on geometry:** Prevent the latent flows from drifting towards **geometrically valid** but **visually meaningless** configurations.

This part formalizes the observation model and its integration into the final variational objective.

6.1 Observation Model and Conditional Independence

We consider a longitudinal sequence of observations:

$$\mathbf{x}_{0:T} = \{x_0, x_1, \ldots, x_T\}, \quad x_t \in \mathcal{X},$$

where $\mathcal{X}$ is the space of medical images (MRI, CT scans, ...).
Each observation x_t is associated with a latent state $z_t \in \mathcal{M}$, where $\mathcal{M}$ is the **Riemannian latent manifold** learned by the model.

Definition 6.1 - Observation Likelihood

We define the **conditional likelihood of an observation** given its latent state as:

$$p_\theta(x_t \mid z_t),$$

parameterized by a **neural decoder** $D_\theta : \mathcal{M} \to \mathcal{X}$, which maps latent coordinates to image space.
In practice, this likelihood is instantiated as:

- **Gaussian model (continuous images):**

$$p_\theta(x_t \mid z_t) = \mathcal{N}(x_t; D_\theta(z_t), \sigma_x^2 I)$$

- Perceptual / hybrid variants for improved visual realism.

Proposition 6.1 — Conditional Independence of Observations

Assume the **longitudinal generative model** defined by

$$p_\theta(x_{0:T}, z_{0:T}) = p(z_{0:T}) \prod_{t=0}^{T} p_\theta(x_t \mid z_t) \tag{6.1}$$

Then the observations are conditionally independent given the latent trajectory $z_{0:T}$, i.e.

$$p_\theta(x_{0:T} \mid z_{0:T}) = \prod_{t=0}^{T} p_\theta(x_t \mid z_t) \tag{6.2}$$

In particular, for any $s \neq t$,

$$x_s \perp\!\!\!\perp x_t \mid z_{0:T}$$

This implies that all temporal correlations in the observed data are **mediated exclusively through** the latent evolution z_t, and not through direct dependencies between the observations themselves.
Each observation is therefore interpreted as an **independent measurement** of the **instantaneous latent physiological state**.

Proof:

Starting from the joint distribution (6.1),

$$p_\theta(x_{0:T}, z_{0:T}) = p(z_{0:T}) \prod_{t=0}^{T} p_\theta(x_t \mid z_t),$$

the conditional distribution of observations given the latent trajectory is obtained via Bayes' rule:

$$p_\theta(x_{0:T} \mid z_{0:T}) = \frac{p_\theta(x_{0:T}, z_{0:T})}{p(z_{0:T})} = \prod_{t=0}^{T} p_\theta(x_t \mid z_t)$$

The product form implies that each observation x_t depends only on its corresponding latent state z_t, and not on other observations x_s for $s \neq t$.
Therefore,

$$x_s \perp\!\!\!\perp x_t \mid z_{0:T}, \quad \forall s \neq t.$$

This establishes conditional independence.

Clinical Interpretation (Longitudinal Medical

Validity and Modeling Scope

Imaging)

In the context of disease progression imaging, this hypothesis carries a precise biological meaning:

- Each scan is assumed to be a **snapshot of the current physiological state** of the patient.
- All temporal dependencies between scans arise from the **evolution of the hidden pathology**, represented by the latent trajectory z_t.
- The imaging process itself is assumed **memoryless**: the scanner does not ***remember*** previous acquisitions.

Concretely:

- If a ***tumor grows***, its shape evolution is encoded in the latent trajectory z_t.
- The images x_t are simply ***noisy measurements*** of this evolving state, not causally linked to each other.

This reflects the clinical reality that MRI or CT machines **do not introduce temporal artifacts that persist across acquisitions** (beyond mild scanner drift or registration noise).

FIND MORE LITTERATURE SOURCES FOR THIS MAYBE!!!

This assumption is **well-justified** when:

- Images are acquired **independently** (standard clinical protocols).
- Temporal correlation is due to **biological evolution** rather than **sensor memory**.
- The objective is physiological modeling of **disease dynamics**.

However, it ***may be violated if***:

- Strong temporal preprocessing is applied (temporal smoothing, video-denoising),
- Motion artefacts accumulate over successive frames,
- Scanner drift introduces correlated noise across time.

In such cases, a more complex observation model of the form

$$p_\theta(x_t \mid x_{t-1}, z_t)$$

could be considered, but this is intentionally avoided in RLVAE to maintain geometric interpretability and tractability.

Conceptual Summary

The modeling choice:

$$p_\theta(x_{0:T} \mid z_{0:T}) = \prod_{t=0}^{T} p_\theta(x_t \mid z_t)$$

Means that:

- The **latent space** captures **disease evolution**.
- The **decoder** captures **imaging physics**.
- The **temporal structure** lies entirely in the **manifold trajectory**.

This clean separation is essential for:

- **Interpretable** disease **trajectories**.
- **Stable** geometric learning.
- Reliable **interpolation** and **prediction** across **time**.

6.2 Reconstruction Term and Data Fidelity Constraint

The reconstruction term represents the data-consistency component of the ELBO. It ensures that the latent trajectory generated by the geometric flows remains faithful to the observed medical image sequence.

Proposition 6.2 - Reconstruction Term in the Longitudinal ELBO

Under the conditional independence hypothesis (**Proposition 6.1**), the **reconstruction term of the ELBO** decomposes as:

$$\mathcal{L}_{rec}(\theta, \phi) = \sum_{t=0}^{T} \mathbb{E}_{q_\phi(z_t \mid x_{0:T})} \left[\log p_\theta(x_t \mid z_t)\right]$$

Equivalently, the **negative** reconstruction loss is given by

$$\mathcal{L}_{rec}^{-}(\theta, \phi) = -\sum_{t=0}^{T} \mathbb{E}_{q_\phi(z_t \mid x_{0:T})} \left[\log p_\theta(x_t \mid z_t)\right]$$

This term enforces that the **decoded trajectory**

$$\hat{x}_t = D_\theta(z_t)$$

remains **close to the observed sequence** $\{x_t\}_{t=0}^{T}$ in image space.

Proof:

Starting from the ELBO expression:

$$\mathcal{L}_{ELBO} = \mathbb{E}_{q_\phi(z_{0:T}|x_{0:T})}\left[\log p_\theta(x_{0:T} \mid z_{0:T})\right] - \mathrm{KL}(q_\phi(z_{0:T}) \| p(z_{0:T})),$$

and using the conditional independence factorization:

$$p_\theta(x_{0:T} \mid z_{0:T}) = \prod_{t=0}^{T} p_\theta(x_t \mid z_t),$$

we obtain:

$$\log p_\theta(x_{0:T} \mid z_{0:T}) = \sum_{t=0}^{T} \log p_\theta(x_t \mid z_t)$$

Therefore,

$$\mathbb{E}_{q_\phi}\left[\log p_\theta(x_{0:T} \mid z_{0:T})\right] = \sum_{t=0}^{T} \mathbb{E}_{q_\phi(z_t|x_{0:T})}\left[\log p_\theta(x_t \mid z_t)\right]$$

This isolates the reconstruction contribution as a sum of per-time-step likelihood terms.

Remark (Geometric Stability):
Although the reconstruction term does **not explicitly depend** on the **Riemannian metric**, it plays a crucial indirect stabilizing role.
It counterbalances the **Expansion Drive** by enforcing the following constraints:

- Large **geometric distortions** that improve prior likelihood but degrade reconstruction are **penalized**.
- The latent flows must **preserve clinically meaningful structure** to minimize visual error.
- The learned geometry must remain **predictive of observable disease patterns**.

Thus, the reconstruction term acts as a **data-anchoring force**, competing with purely geometric incentives.

6.3 Choice of Likelihood and Reconstruction Metric

The reconstruction term introduced in **Section 6.2** depends entirely on the choice of likelihood model

$$p_\theta(x_t \mid z_t),$$

which determines how similarity between generated and observed images is measured.
This choice defines the geometry of the data space and directly influences gradient structure, stability, and perceptual fidelity.

Definition 6.2 - Parametric Likelihood Model

Let $D_\theta : \mathcal{M} \to \mathcal{X}$ be the decoder mapping latent coordinates z_t to image space.
We define the conditional likelihood as:

$$p_\theta(x_t \mid z_t) = \mathcal{P}(x_t; D_\theta(z_t), \Sigma_x),$$

where $\mathcal{P}$ denotes a parametric family and Σ_x encodes the observation noise model.

6.3.1 Gaussian Likelihood (Quadratic Metric)

The most common choice is the **Gaussian likelihood**:

$$p_\theta(x_t \mid z_t) = \mathcal{N}\left(x_t; D_\theta(z_t), \sigma_x^2 I\right)$$

This yields the reconstruction loss:

$$\log p_\theta(x_t \mid z_t) = \frac{1}{2\sigma_x^2}\|x_t - D_\theta(z_t)\|_2^2 + C,$$

6.3.2 Bernoulli Likelihood (Logistic Metric)

For normalized or binary images:

$$p_\theta(x_t \mid z_t) = \prod_{i=1}^{n} \mathrm{Bernoulli}(x_{t,i}; \sigma(D_\theta(z_t)_i))$$

leading to:

$$\log p_\theta(x_t \mid z_t) = \mathrm{BCE}(x_t, D_\theta(z_t)).$$

equivalent to a **weighted mean-squared error**.

This introduces a **cross-entropy reconstruction geometry**.

Properties:

- Induces a **Euclidean metric** in image space.
- Differentiable, stable, and convex in x_t.
- Equivalent to maximum likelihood under i.i.d. Gaussian noise.

6.3.3 Perceptual Likelihood (Feature-Space Geometry)

A more expressive alternative introduces a feature embedding $\varphi : \mathcal{X} \to \mathcal{F}$, leading to:

$$p_\theta(x_t \mid z_t) \propto \exp\left(-\frac{1}{2\sigma_p^2}\|\varphi(x_t) - \varphi(D_\theta(z_t))\|_2^2\right)$$

This corresponds to a **Gaussian likelihood in feature space**.

6.3.4 Hybrid Likelihoods

A general formulation combines pixel and perceptual metrics:

$$\mathcal{L}_{rec}^t = \lambda_{pix}\|x_t - D_\theta(z_t)\|_2^2 + \lambda_{perc}\|\varphi(x_t) - \varphi(D_\theta(z_t))\|_2^2$$

This defines a composite metric on $\mathcal{X}$.

Proposition 6.3 - Influence of Likelihood on Latent Geometry

The choice of likelihood $p_\theta(x_t \mid z_t)$ induces an **implicit metric** on the latent manifold through the **pullback**:

$$g_{latent}(z_t) = J_{D_\theta}(z_t)^\top \, g_{obs} \, J_{D_\theta}(z_t),$$

where g_{obs} is the metric defined by the reconstruction loss in observation space.
Thus, the reconstruction model defines **not** **merely visual fidelity**, but the effective geometry of the learned latent manifold.

Proof:

The reconstruction loss defines a distance function $d(x, \hat{x})$.

Since $x = D_\theta(z)$, the differential of this distance induces a metric on $\mathcal{M}$ by pullback:

$$\langle u, v\rangle_{g_z} = \langle J_D(z)u, J_D(z)v\rangle_{g_{obs}}$$

This corresponds exactly to the stated expression.

6.3.5 Summary

Likelihood	Reconstruction Geometry	Key Effect
Gaussian	Euclidean (MSE)	Smooth, simple, stable
Bernoulli	Cross-Entropy	Discrete modeling
Perceptual	Feature geometry	Structure-aware fidelity
Hybrid	Multi-scale geometry	Balanced constraints

The **likelihood choice** therefore determines:

- The **shape** of reconstruction basins.
- The **smoothness** of gradients.
- The interaction between **decoder** and **latent geometry**.

6.4 Practical Reconstruction Loss Used in RLVAE

While Section 6.3 presented several admissible likelihood families, the operational implementation of RLVAE uses a concrete reconstruction objective chosen for numerical stability and compatibility with geometric learning.

Definition 6.3 - **Operational Reconstruction Loss**

The **reconstruction loss** at time t is defined as:

$$\mathcal{L}_{rec}^{(t)} = -\mathbb{E}_{q_\phi(z_t|\mathbf{x})}\left[\log p_\theta(x_t \mid z_t)\right]$$

In practice, we adopt a **Gaussian likelihood with fixed variance**:

$$p_\theta(x_t \mid z_t) = \mathcal{N}(x_t; D_\theta(z_t), \sigma_x^2 I),$$

leading to the explicit loss:

$$\mathcal{L}_{rec}^{(t)} = \frac{1}{2\sigma_x^2}\|x_t - D_\theta(z_t)\|_2^2 + C$$

The total reconstruction loss is:

$$\mathcal{L}_{rec} = \sum_{t=0}^{T} \mathcal{L}_{rec}^{(t)}$$

Proposition 6.4 - **Temporal Consistency Constraint**

Although the likelihood is **factorized across time**, the reconstruction loss enforces temporal consistency implicitly through the **deterministic trajectory**:

$$z_{t+1} = f_{t+1}(z_t),$$

so that **reconstruction accuracy** is jointly constrained by the **entire latent path**.

Proof:

Each term $\|x_t - D_\theta(z_t)\|^2$ depends on z_t, which itself depends on z_0 via the flow composition.
Therefore minimizing reconstruction error at time t propagates constraints backward through the entire trajectory.

6.5 Interaction Between Reconstruction and Latent Geometry

The reconstruction term **does not** operate **independently of geometry**, it induces biases in the learned metric structure.

Proposition 6.5 - Geometric Alignment through Reconstruction

Let the decoder be a smooth map

$$D_\theta : \mathcal{M} \to \mathcal{X}, \quad z \mapsto \mu(z)$$

and let the likelihood be of the form

$$p_\theta(x \mid z) = p(x \mid \mu(z)),$$

where $p(\cdot \mid \mu)$ is a parametric family (e.g. Gaussian with mean μ) endowed with a Fisher information metric g_{obs} on the parameter space of μ.
Then the **Fisher information metric in latent space** induced by the reconstruction term is the **pullback** of g_{obs} by the decoder:

$$g_{\text{latent}}(z) = J_{D_\theta}(z)^\top \, g_{\text{obs}}(\mu(z)) \, J_{D_\theta}(z),$$

where $J_{D_\theta}(z) = \frac{\partial \mu(z)}{\partial z}$ is the Jacobian of the decoder.
In particular, minimizing the expected reconstruction loss is equivalent (at second order) to **penalizing deviations** measured in this pullback metric, hence **implicitly** regularising the latent geometry toward alignments that minimize this distortion.

Proof:
We make the link via the Fisher information of the likelihood with respect to the latent variable z.

1. **Set-up:**
   Assume the likelihood factorises through a decoder:

   $$p_\theta(x \mid z) = p(x \mid \mu(z)),$$

   where $\mu(z) = D_\theta(z)$ and $p(x \mid \mu)$ is a smooth family of densities on $\mathcal{X}$.
   Define the **log-likelihood**:

   $$\ell(x, z) \;= \log p(x \mid \mu(z))$$

   We consider the Fisher information **in z-coordinates**, defined as:

   $$g_{\text{latent}}(z) \;= \mathbb{E}_{x \sim p(\cdot \mid \mu(z))} \left[\nabla_z \ell(x, z) \, \nabla_z \ell(x, z)^\top \right]$$

   This is the canonical Riemannian metric associated with maximum likelihood / reconstruction at point z.
2. **Chain rule: gradient with respect to z:**
   By the chain rule:

   $$\nabla_z \ell(x, z) = J_{D_\theta}(z)^\top \, \nabla_\mu \log p(x \mid \mu(z))$$

   Here:

   - $J_{D_\theta}(z) \in \mathbb{R}^{d_x \times d_z}$ is the Jacobian of the decoder w.r.t. z,
   - $\nabla_\mu \log p(x \mid \mu)$ is the gradient of the log-likelihood w.r.t. its natural parameter μ.
3. **Plug into Fisher information in latent space:**
   We now plug this into the definition of g_{latent}:

   $$\begin{aligned} g_{\text{latent}}(z) &= \mathbb{E}_{x \sim p(\cdot \mid \mu(z))} \left[\left(J_{D_\theta}(z)^\top \nabla_\mu \log p(x \mid \mu(z)) \right) \left(J_{D_\theta}(z)^\top \nabla_\mu \log p(x \mid \mu(z)) \right)^\top \right] \\ &= J_{D_\theta}(z)^\top \, \mathbb{E}_{x \sim p(\cdot \mid \mu(z))} \left[\nabla_\mu \log p(x \mid \mu(z)) \, \nabla_\mu \log p(x \mid \mu(z))^\top \right] \; J_{D_\theta}(z) \end{aligned}$$

   Define the Fisher information **in the observation-parameter space**:

   $$g_{\text{obs}}(\mu) \;= \mathbb{E}_{x \sim p(\cdot \mid \mu)} \left[\nabla_\mu \log p(x \mid \mu) \, \nabla_\mu \log p(x \mid \mu)^\top \right]$$

   Then the expression above becomes exactly:

   $$g_{\text{latent}}(z) = J_{D_\theta}(z)^\top \, g_{\text{obs}}(\mu(z)) \, J_{D_\theta}(z)$$

This is **precisely the pullback** of the metric g_{obs} by the map D_θ.

4. **Link with reconstruction loss:**
The reconstruction loss is the **negative log-likelihood**:

$$\mathcal{L}_{rec}(z) = -\mathbb{E}_{x\sim p(\cdot|\mu(z))}[\ell(x, z)]$$

The **second-order expansion** of $\mathcal{L}_{rec}$ around a point z involves exactly the Fisher information metric:

$$\nabla_z^2 \mathcal{L}_{rec}(z) \approx g_{\text{latent}}(z),$$

in the usual Fisher/Natural gradient approximation (the Hessian of the negative log-likelihood equals the Fisher, up to regularity conditions).
Thus:

- Local perturbations δz are penalised (at second order) by the quadratic form

$$\delta z^\top g_{\text{latent}}(z)\delta z$$

- Since g_{latent} is the pullback metric $J_D^\top g_{\text{obs}} J_D$, the reconstruction term **implicitly discourages** latent geometries in which the decoder strongly distorts the observation-space metric.

Interpretation (strictly mathematical)

- Latent directions **poorly mapped** by the **decoder** receive **higher gradient penalties**.
- The decoder constrains **latent curvature** indirectly through sensitivity structure.
- Regions of **high curvature in latent space** correspond to regions of decoder nonlinearity.

Lemma 6.1 - Local Conditioning Effect

The reconstruction Hessian at z satisfies:

$$H_{rec}(z) \approx J_D(z)^\top J_D(z),$$

inducing **anisotropy** in the latent space ***aligned*** with the **decoder Jacobian**.

6.6 Likelihood-Induced Curvature and Flow Interaction

We now connect reconstruction with flow geometry.

Proposition 6.6 - Reconstruction-Induced Curvature

Let $z_t = F_t(z_0)$.
Then the reconstruction loss induces an effective curvature tensor in the latent manifold:

$$\tilde{G}(z_0) = \sum_{t=0}^{T} J_{F_t}(z_0)^\top J_{D_\theta}(z_t)^\top J_{D_\theta}(z_t) J_{F_t}(z_0)$$

This **curvature** penalizes **flow directions** that create **large reconstruction distortion**.

Proof (Sketch):

Each reconstruction term contributes a quadratic form:

$$\|x_t - D(z_t)\|^2 \approx \delta z_0^\top J_{F_t}^\top J_D^\top J_D J_{F_t} \delta z_0$$

Summing over t yields the expression.

Corollary 6.1 - Coupling with Flow Regularization

The total geometric force acting on the flow is:

$$\mathcal{F}_{flow} = (\lambda_{flow} - 2)\log|\det J_F| + \mathrm{Tr}(\tilde{G}(z_0))$$

This defines a joint tension between:

- **Volume** distortion.
- **Reconstruction-induced** curvature.

Remark (Metric Interaction):
The reconstruction loss implicitly introduces a **secondary Riemannian structure** on the latent space through the decoder pullback metric.
This metric **may not coincide** with the intrinsic manifold metric G learned from population geometry.
The learned representation therefore results from a **compromise** between statistical geometry and observation geometry, providing a geometrically consistent yet perceptually faithful embedding.

Appendix A — Explicit Form of the Base Proposal Distribution

For completeness and full transparency of the variational family, we detail here the analytical expression of the base proposal distribution $q_0(z_0|x)$.

Recall that the encoder outputs a latent mean $\mu(x) \in \mathbb{R}^d$, and that the covariance matrix is chosen to respect the local geometry through:

$$\Sigma_\mu(x) = \alpha\, G(\mu(x)),$$

where $\alpha > 0$ is a scalar hyperparameter controlling the global uncertainty scale and $G(\mu(x))$ is the Riemannian metric evaluated at the encoder mean.

The base proposal is therefore:

$$q_0(z_0|x) = \mathcal{N}\big(z_0; \mu(x), \alpha G(\mu(x))\big)$$

The log-density admits the closed-form expression:

$$\log q_0(z_0|x) = -\frac{d}{2}\log(2\pi) - \frac{1}{2}\log\det(\alpha G(\mu(x))) - \frac{1}{2}(z_0 - \mu(x))^\top(\alpha G(\mu(x)))^{-1}(z_0 - \mu(x))$$

Using determinant and inverse identities, this simplifies to:

$$\log q_0(z_0|x) = -\frac{d}{2}\log(2\pi) - \frac{d}{2}\log\alpha - \frac{1}{2}\log\det G(\mu(x)) - \frac{1}{2\alpha}(z_0 - \mu(x))^\top G(\mu(x))^{-1}(z_0 - \mu(x)).$$

This expression makes explicit the geometric structure of the proposal:

- The quadratic term $(z_0 - \mu(x))^\top G(\mu(x))^{-1}(z_0 - \mu(x))$ corresponds to a Mahalanobis distance weighted by the local precision field, meaning deviations from $\mu(x)$ are measured in the intrinsic geometry of the learned manifold.
- The entropy correction term $\frac{1}{2}\log\det G(\mu(x))$ introduces spatially varying uncertainty depending on the local geometry, reflecting non-uniform latent density.
- The scalar α decouples global uncertainty from geometric structure, allowing fine control over the sharpness of the posterior independently of the metric.

Therefore, the base proposal uncertainty is not isotropic, but adapts to the learned geometry:

- high precision regions $G^{-1}(\mu(x)) \gg I$ induce tighter distributions,
- low precision regions allow broader exploration.

This explicit decomposition clarifies how the encoder's epistemic uncertainty is intrinsically modulated by the Riemannian structure, serving as the initial stabilised estimate before RHMC geometric refinement.

PART VIII — Implementation and Numerical Stabilisation

8.1 Algorithmic Pseudocode

- Full forward + KL estimation pipeline

8.2 Stability Mechanisms

- SPD enforcement
- Cholesky safety
- Δ_{kin} and Δ_{vol} monitoring

8.3 Hyperparameter Sensitivity

- RHMC steps
- Step size
- Metric temperature

PART IX — Diagnostic & Failure Mode Analysis

9.1 Expected Pathologies

- μ drift
- Metric collapse
- Exploding condition numbers

9.2 Debugging Protocol

- How to detect bias sources
- Interpretation of gradient logs

PART X — Extensions & Perspectives

10.1 Toward Dynamic Metrics

10.2 Comparison with RHVAE / Riemannian Flows

10.3 Generalisation to Real Medical Data
